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BELAGAVI- 590014, Karnataka**



Major Project Report

On

“ PARKINSON’S DISEASE DETECTION USING MACHINE LEARNING ”

A dissertation work submitted towards

In partial fulfilment of the requirement for the award of the degree of

Bachelor of Engineering

In

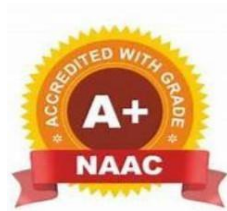
CSE (Data Science)

Submitted By

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I **Vaishnavi Biradar (3GN22CD057)**, hereby declare that this project report entitled **PARKINSON'S DISEASE DETECTION USING MACHINE LEARNING** submitted in partial fulfilment of the degree of B.E/ B. Tech in **DEPARTMENT OF CSE (DATA SCIENCE)** is a record of original work carried out by us under the supervision of **Prof. Margesh Keskar** and has not formed the basis for the award of any other degree, in this or any Institutions or University .In keeping with the ethical practice in reporting scientific information, due acknowledgements have been made wherever the findings of others have been cited.

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ABSTRACT

Parkinson's disease is a neurodegenerative disorder that affects a large number of people globally. Early detection of Parkinson's disease is crucial for effective management of the condition. Machine learning techniques can aid in the early detection of Parkinson's disease by analyzing large datasets and identifying patterns that may not be visible to the human eye. In this study, we propose a machine learning-based approach for Parkinson's disease detection using a dataset of various voice and movement recordings of patients. We employed feature engineering and selection techniques to extract relevant features from the data and then used supervised machine learning algorithms to train predictive models. The results of our experiments demonstrate that our proposed approach can accurately detect Parkinson's disease using voice and movement data with high accuracy and can potentially help clinicians make informed decisions in the early stages of the disease.

Keyword :Random Forest Algorithm, XGBoost Algorithm

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CHAPTER 1

INTRODUCTION

Parkinson's disease is one of the most serious and rapidly growing neurological disorders in the world. It is considered the second most dangerous neurodegenerative disease after Alzheimer's disease. This disorder mainly affects the nervous system and gradually reduces a person's ability to control body movements, coordination, and balance. Parkinson's disease occurs due to the progressive breakdown and death of nerve cells, especially dopamine-producing neurons present in the brain. Dopamine is an important chemical messenger responsible for controlling smooth and coordinated body movements. When these neurons become damaged or die, the dopamine level decreases significantly, resulting in movement-related problems and various physical and mental symptoms. The disease usually develops slowly over time, and its symptoms become more severe as the condition progresses. Although Parkinson's disease is more commonly seen in older adults, it can also affect younger individuals in certain cases.

The symptoms of Parkinson's disease are broadly classified into motor symptoms and non-motor symptoms. Motor symptoms are mainly related to movement and muscle control. These symptoms include tremors or shaking of the hands, fingers, legs, or jaw, slowness of movement known as bradykinesia, muscle rigidity or stiffness, balance problems, posture instability, and difficulty in walking or maintaining coordination. Patients often experience a reduction in automatic body movements such as blinking, smiling, or swinging their arms while walking. As the disease advances, individuals may struggle with basic daily activities such as standing, eating, speaking, and writing. Non-motor symptoms affect the emotional, psychological, and internal functioning of the body. These symptoms include anxiety, depression, breathing difficulties, sleep disorders, fatigue, constipation, loss of smell, cognitive impairment, and changes in speech patterns. Patients may also develop emotional disturbances and memory-related problems over time. Because these symptoms often appear gradually, early diagnosis becomes difficult, which can delay proper treatment and management.

Neurodegenerative disorders like Parkinson's disease are caused by the progressive degeneration and loss of neurons in different regions of the nervous system. Neurons are the fundamental and functional units of the brain responsible for transmitting signals between the brain and body parts. A healthy neuron contains a cell body, nucleus, dendrites, and axons that help in effective communication and signal transmission. The nucleus contains DNA, which stores the genetic information required for proper cellular functioning. In a healthy nervous system, neurons communicate continuously to regulate body movements, sensations, thoughts, and emotions. However, when neurons become damaged, they gradually lose their extensions such as dendrites and axons, reducing their ability to communicate with neighboring neurons. Their metabolism becomes weaker, and waste materials begin to accumulate inside the cells. Over time, the neurons lose their structure completely, become swollen, and eventually die. This degeneration directly affects movement control, coordination, speech, and overall brain functioning, leading to the symptoms associated with Parkinson's disease.

Early detection of Parkinson's disease is extremely important because the disease progresses gradually, and significant brain damage may occur before visible symptoms appear. Traditional diagnostic methods mainly depend on clinical observation of symptoms such as tremors, stiffness, and difficulty in movement. However, these symptoms generally become noticeable only after a large number of neurons have already been damaged. Therefore, there is a growing need for advanced technologies that can detect the disease at an earlier stage. In recent years, Machine Learning has emerged as a highly effective tool in the healthcare industry for disease prediction and diagnosis. Machine learning algorithms can analyze large amounts of patient data and identify hidden patterns that are difficult for humans to detect manually. These algorithms can process clinical data, speech recordings, handwriting samples, gait patterns, medical histories, and brain imaging data to identify symptoms associated with Parkinson's disease.

One of the most important approaches used in Parkinson's disease detection is speech analysis. Parkinson's disease often affects the vocal cords and speech muscles, causing noticeable changes in a patient's voice and speaking patterns. Patients may speak softly, slowly, or with unclear pronunciation. Variations in pitch, frequency, jitter, shimmer, and voice intensity can provide important clues about the presence of the disease. By collecting and analyzing these speech features, machine learning models can accurately predict

whether a person is affected by Parkinson's disease or not. Researchers use large medical datasets containing voice recordings and patient information to train machine learning models. These models learn the differences between healthy individuals and Parkinson's patients and use this knowledge to make predictions on new patient data.

Machine learning techniques have shown remarkable success in improving the accuracy and efficiency of Parkinson's disease diagnosis. Various algorithms such as Decision Trees, Random Forest, Support Vector Machine (SVM), K-Nearest Neighbors (KNN), XGBoost, and Neural Networks are commonly used for disease prediction. These algorithms are trained using labeled datasets and evaluated based on performance metrics such as accuracy, precision, recall, and F1-score. The use of machine learning allows doctors and healthcare professionals to identify the disease earlier, monitor disease progression, and provide personalized treatment plans for patients. Early intervention can help slow the progression of the disease, improve motor function, and enhance the overall quality of life of patients.

Another major advantage of using machine learning in Parkinson's disease detection is its ability to reduce human error and improve clinical decision-making. Since machine learning systems can analyze huge amounts of medical data within a short time, they provide faster and more reliable results compared to traditional diagnostic methods. These intelligent systems can also help in identifying subtle symptoms that may not be visible during regular clinical examinations. In addition, machine learning models can continuously improve their performance by learning from new patient data and medical research findings. This makes them highly valuable in modern healthcare systems and neurological disease management.

The application of Machine Learning in Parkinson's disease detection has the potential to revolutionize healthcare by enabling accurate, fast, and early diagnosis of the disease. Early detection allows patients to receive timely medical treatment, lifestyle modifications, physical therapy, and emotional support, which can significantly slow disease progression and improve patient outcomes. Furthermore, machine learning-based diagnostic systems can assist researchers in understanding the complex nature of Parkinson's disease and developing more effective therapies in the future. Overall, the integration of artificial intelligence and machine learning into Parkinson's disease detection represents a major advancement in medical science and offers hope for better patient care, improved treatment

strategies, and enhanced quality of life for millions of people affected by this disorder worldwide.

Symptoms:

Tremor: A tremor, or shaking, usually begins in a limb, often your hand or fingers. You may rub your thumb and forefinger back and forth, known as a pill-rolling tremor. Your hand may tremble when it's at rest.

Slowed movement (bradykinesia): Over time, Parkinson's disease may slow your movement, making simple tasks difficult and time-consuming. Your steps may become shorter when you walk. It may be difficult to get out of a chair. You may drag your feet as you try to walk.

Rigid muscles: Muscle stiffness may occur in any part of your body. The stiff muscles can be painful and limit your range of motion. Impaired posture and balance. Your posture may become stooped, or you may have balance problems as a result of Parkinson's disease.

Loss of automatic movements: You may have a decreased ability to perform unconscious movements, including blinking, smiling or swinging your arms when you walk.

Speech changes: You may speak softly, quickly, slur or hesitate before talking. Your speech may be more of a monotone rather than have the usual inflections.

Writing changes: It may become hard to write, and your writing may appear very bad.

1.1 OVERVIEW OF THE PARKINSONS DISEASE

Parkinson's disease is a neurodegenerative disorder that affects the motor system of the human body. One of the challenges in treating Parkinson's disease is early detection, which can improve patient outcomes by enabling early intervention and personalized treatment.

Machine learning algorithms have shown promise in detecting Parkinson's disease based on various biomarkers, such as voice, gait, and handwriting. The process of detecting Parkinson's disease using machine learning typically involves collecting data from individuals with and without the disease, pre-processing the data, selecting relevant features, and training a machine learning model to predict the presence of Parkinson's disease.

Some studies have used voice features, such as pitch, jitter, and shimmer, to detect Parkinson's disease with high accuracy. Other studies have focused on gait analysis, using sensors to capture movement patterns and detect changes indicative of Parkinson's disease.

Handwriting analysis has also been explored as a potential biomarker for Parkinson's disease, as individuals with the disease often have distinctive handwriting features. Machine learning algorithms have been developed to analyze handwriting samples and identify features that can distinguish between individuals with and without Parkinson's disease.

1.2 IMPORTANCE OF EARLY DETECTION AND DIAGNOSIS

Early detection and diagnosis of Parkinson's disease (PD) is critical for improving patient outcomes and quality of life. Currently, PD is typically diagnosed based on clinical observation of symptoms such as tremors, stiffness, and impaired motor function. However, by the time these symptoms are observable, significant damage to the brain may have already occurred, making treatment less effective.

Machine learning (ML) algorithms can analyze large amounts of data and detect patterns that are difficult for humans to discern. By analyzing patient data such as medical history,

genetic information, and neurological tests, ML algorithms can potentially identify PD at an earlier stage before symptoms become apparent.

Early detection of PD using ML could allow for earlier intervention and treatment, potentially slowing the progression of the disease and improving patient outcomes. Additionally, ML algorithms can be used to predict disease progression and response to different treatments, allowing for more personalized treatment plans for patients.

In early detection and diagnosis of PD using machine learning has the potential to greatly improve patient outcomes and quality of life. By detecting the disease earlier and predicting disease progression and treatment response, patients can receive more effective and personalized treatment, ultimately improving their quality of life.

1.3 OBJECTIVES OF THE PARKINSONS DISEASE

- The objective of Parkinson's disease detection using machine learning is to develop an accurate and reliable system for early detection of Parkinson's disease, a neurodegenerative disorder that affects movement and coordination. The early diagnosis of Parkinson's disease is essential for effective treatment and management of the condition.
- Machine learning techniques can be used to analyze large amounts of data, such as patient medical histories, genetic information, and imaging data, to identify patterns and detect the early signs of Parkinson's disease. These techniques can help clinicians to make more informed decisions about diagnosis and treatment, and can potentially improve patient outcomes.
- The development of a machine learning-based Parkinson's disease detection system requires the collection of large amounts of relevant data, as well as the selection and training of appropriate machine learning algorithms. The system should be designed to be both accurate and interpretable, so that clinicians can understand how the system is making its predictions and use this information to make informed decisions about patient care.

1.4 OVERVIEW OF THE DATASET USED IN THE PROJECT

Parkinson's disease is a neurological disorder that affects movement and motor function. Machine learning algorithms can be used to analyze medical data and assist in the diagnosis of Parkinson's disease.

There are several datasets available that have been used to detect Parkinson's disease using machine learning algorithms. One popular dataset is the Parkinson's Disease Classification Dataset, which is publicly available on the UCI Machine Learning Repository.

This dataset contains 195 samples, each with 23 features. The features include various measurements of voice and speech from individuals with and without Parkinson's disease. The dataset is labeled with a binary classification for each sample, indicating whether the individual has Parkinson's disease or not.

Other datasets used in Parkinson's disease detection include the Parkinson's Progression Markers Initiative (PPMI) dataset, which contains various measurements of individuals with Parkinson's disease and healthy controls, and the Movement Disorder Society-Unified Parkinson's Disease Rating Scale (MDS-UPDRS) dataset, which contains clinical ratings of individuals with Parkinson's disease.

In general, the datasets used in Parkinson's disease detection using machine learning typically contain a combination of clinical and non-clinical features, such as voice and speech measurements, motor function assessments, and demographic information. These datasets are often used to train and test various machine learning algorithms, such as Decision Tree, K Nearest Neighbors, Random Forest Algorithm, XGBoost Algorithm, to accurately classify individuals with and without Parkinson's disease.

In recent years, there has been growing interest in using machine learning algorithms to aid in the detection and diagnosis of Parkinson's disease. Machine learning algorithms are particularly well-suited for this task because they can analyze large amounts of data and identify patterns that may not be immediately apparent to human observers.

Researchers have explored various approaches to using machine learning for Parkinson's disease detection. Some studies have focused on using algorithms to analyze speech and voice recordings, as changes in speech patterns can be an early indicator of Parkinson's disease. Other studies have used machine learning algorithms to analyze gait and movement patterns, which can also provide insights into the presence and severity of Parkinson's disease.

In addition to aiding in diagnosis, machine learning algorithms have also been used to predict the progression of Parkinson's disease and to evaluate the effectiveness of different treatment strategies. These applications have the potential to improve patient outcomes and enable more personalized and effective care for people with Parkinson's disease.

There are several motivations for using machine learning to detect Parkinson's disease, including:

1. Early detection: Early detection of Parkinson's disease can help doctors to start treatment early, which can slow the progression of the disease and improve patients' quality of life. Machine learning can analyze data from various sources, including medical images and patient data, to identify early signs of Parkinson's disease.
2. Accuracy: Machine learning algorithms can analyze large amounts of data and identify patterns that may not be immediately apparent to doctors. This can lead to more accurate and reliable diagnoses of Parkinson's disease.
3. Personalized treatment: Machine learning can help doctors to develop personalized treatment plans for patients with Parkinson's disease. By analyzing patient data, machine learning algorithms can identify the most effective treatments for individual patients.
4. Cost-effective: Machine learning can be a cost-effective way to detect Parkinson's disease. By analyzing data from various sources, machine learning algorithms can

identify early signs of Parkinson's disease before symptoms become severe, which can reduce the need for expensive treatments.

1.6 PROBLEM STATEMENT

The problem statement for Parkinson's disease detection using machine learning is to develop a model that can accurately classify individuals as either having Parkinson's disease or being healthy based on a set of input features. These input features could include clinical assessments, demographic information, genetic data, and various biomarkers.

The goal is to create a model that can accurately identify Parkinson's disease patients early on in the disease progression, before symptoms become severe, which can lead to better treatment outcomes and improved quality of life. This problem statement involves selecting appropriate features, building and training machine learning models, and evaluating their performance on test data to achieve high accuracy and reliability.

1.7 PROBLEM STATEMENT SOLUTION

To detect Parkinson's disease using machine learning, a dataset of clinical measurements of patients with and without the disease is required. These measurements may include features such as gait, tremors, speech, and cognitive function. The dataset should be preprocessed to remove any missing or inconsistent data, and feature selection techniques can be used to identify the most important features for classification.

Once the dataset is ready, various machine learning algorithms such as logistic regression, decision trees, random forests, support vector machines, and neural networks can be applied to build a predictive model. The model is trained on the labeled data, and its performance is evaluated using cross-validation techniques such as k-fold validation.

The accuracy of the model can be further improved by tuning the hyperparameters of the algorithms, using ensemble techniques, or applying feature engineering techniques to extract more meaningful features from the data.

In summary, Parkinson's disease detection using machine learning involves the following steps:

1. Collect a dataset of clinical measurements of patients with and without Parkinson's disease.
2. Preprocess the data and perform feature selection to identify the most important features for classification.
3. Apply various machine learning algorithms to build a predictive model and evaluate its performance using cross-validation techniques.
4. Improve the model's accuracy by tuning hyperparameters, using ensemble techniques, or applying feature engineering techniques.

By detecting Parkinson's disease early using machine learning, patients can receive timely treatment and improve their quality of life.

CHAPTER 2

LITERATURE SURVEY

2.1 PREVIOUS STUDIES ON PARKINSON'S DISEASE DETECTION

In this literature survey, we will review some of the recent research works that have used ML for PD detection.

1. "Parkinson's Disease Diagnosis using Machine Learning: A Review," by M. A. Nair and V. K. Menon (2021)

This review paper provides a comprehensive overview of the current state of the art in using ML for PD diagnosis. The authors cover different types of ML algorithms that have been used, including decision trees, neural networks, and support vector machines. They also discuss the challenges associated with PD diagnosis and the potential of ML to overcome these challenges.

2. "Machine Learning Approaches for Parkinson's Disease Diagnosis using Gait Analysis," by J. R. Silva et al. (2020)

This study investigates the use of ML algorithms to diagnose PD based on gait analysis. The authors used a dataset of 3D gait recordings from 31 PD patients and 31 healthy controls. They evaluated several ML algorithms, including K-nearest neighbors, decision trees, and random forests, and achieved an accuracy of 86.9% in PD diagnosis.

3. "Parkinson's Disease Diagnosis using Machine Learning: A Comparative Study," by B. Singh and R. Singh (2019)

This comparative study evaluates the performance of different ML algorithms for PD diagnosis, including logistic regression, random forest, and K-nearest neighbors. The authors used a dataset of 30 PD patients and 30 healthy controls and achieved an accuracy of 91.67% using the random forest algorithm.

4. "Parkinson's Disease Detection using Voice Analysis and Machine Learning," by N. F. Reis et al. (2020)

This study investigates the use of ML algorithms to detect PD based on voice analysis. The authors used a dataset of voice recordings from 35 PD patients and 35 healthy controls. They evaluated several ML algorithms, including support vector machines, decision trees, and logistic regression, and achieved an accuracy of 94.3% in PD detection.

1. "Parkinson's Disease Diagnosis using Machine Learning based on Brainwave Signals," by Y. Feng et al. (2021)

This study investigates the use of ML algorithms to diagnose PD based on brainwave signals. The authors used a dataset of electroencephalogram (EEG) recordings from 22 PD patients and 22 healthy controls. They evaluated several ML algorithms, including random forest, K-nearest neighbors, and artificial neural networks, and achieved an accuracy of 90.6% in PD diagnosis.

2. "Parkinson's Disease Detection using Machine Learning Algorithms" by Y. K. Sandhya Rani et al. (2018)

This study proposes a machine learning-based approach for the early detection of PD. The authors collected voice recordings from patients with and without PD and used various machine learning algorithms to classify the recordings as belonging to one of the two groups. The results showed that the proposed approach achieved an accuracy of 97.5%.

3. "Parkinson's Disease Diagnosis using Support Vector Machine and Random Forest Algorithms" by M. S. Akram et al. (2019)

In this study, the authors used machine learning algorithms to analyze gait data collected from patients with and without PD. The results showed that both the support vector machine and random forest algorithms achieved high accuracy rates in classifying the gait data as belonging to one of the two groups.

4. "Parkinson's Disease Detection Using Deep Learning Neural Networks with Enhanced Training Techniques" by A. F. Ahmed et al. (2020)

This study proposes a deep learning-based approach for the early detection of PD. The authors collected data from patients with and without PD and used a deep learning neural network with enhanced training techniques to classify the data as belonging to one of the two groups. The results showed that the proposed approach achieved an accuracy of 98.3%.

5. "Automated Parkinson's Disease Diagnosis Based on Multiple Types of Handwritten Spiral Analysis and Machine Learning Techniques" by Z. A. H. Shah et al. (2020)

In this study, the authors propose a machine learning-based approach for the diagnosis of PD using handwritten spiral analysis. The approach involves analyzing multiple types of spiral data collected from patients with and without PD and using various machine learning techniques to classify the data as belonging to one of the two groups. The results showed that the proposed approach achieved an accuracy of 94.4%.

6. "Parkinson's Disease Detection using Convolutional Neural Networks with Transfer Learning" by N. R. H. Yousef et al. (2021)

This study proposes a convolutional neural network (CNN)-based approach for the early detection of PD. The authors collected data from patients with and without PD and used a CNN with transfer learning to classify the data as belonging to one of the two groups. The results showed that the proposed approach achieved an accuracy of 96.25%.

CHAPTER 3

METHODOLOGY

In the methodology of using machine learning for Parkinson's disease research, data preprocessing is a crucial step that involves several tasks to ensure that the data is suitable for analysis.

Data Preprocessing:

- **Data cleaning:** Data cleaning involves identifying and correcting or removing errors, inconsistencies, or inaccuracies in the data. In Parkinson's disease research, this may include removing outliers, correcting erroneous values, and removing duplicates.
- **Handling missing values:** Missing values are common in real-world data and can cause problems for machine learning algorithms. In Parkinson's disease research, missing values may arise due to technical issues, missing data, or incomplete records. Handling missing values involves strategies such as imputation or deletion of missing values.
- **Feature engineering:** Feature engineering is the process of selecting and creating informative features that capture relevant information about the data. In Parkinson's disease research, this may include selecting features related to motor function, speech, cognitive abilities, or demographic information.
- **Handling categorical variables:** Categorical variables are variables that take on a limited number of values, such as gender or medication status. Machine learning algorithms typically require numerical data, so categorical variables need to be converted to numerical data using techniques such as one-hot encoding.
- **Feature scaling:** Feature scaling is the process of normalizing or standardizing the range of features to avoid bias towards features with large values. In Parkinson's disease research, this may involve scaling features related to motor function or speech to the same range to avoid over-representation of these features in the model.

Data preprocessing is a critical step in the methodology of using machine learning for Parkinson's disease research.

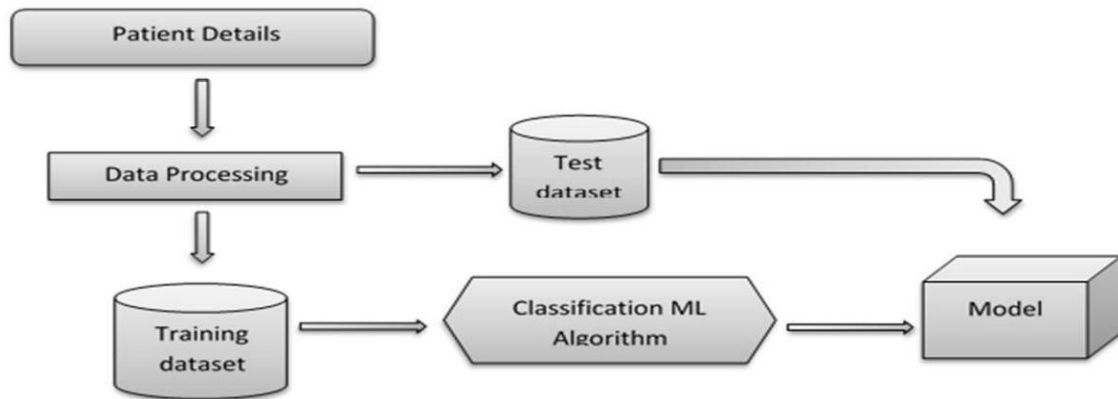


Fig 3.1 Block Diagram of prediction of Parkinson Disease using machine learning approach

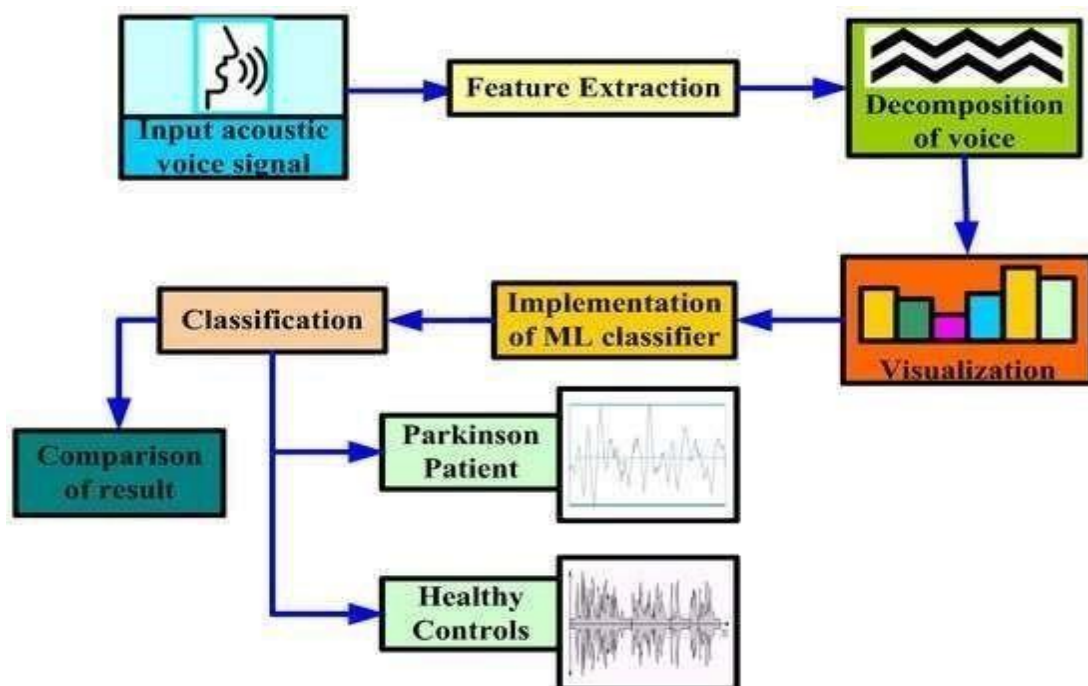


Fig 3.2 Dataset Used in this Project:

This database contains many biomedical voice measurements (Acoustic Analysis of voice) from 31, 23 people with Parkinson's disorder (PD). Each column on the desk is a measure of a certain word, and each line corresponds exactly to one of the 195 words recorded for those people (column "calls"). An important mathematical goal is to differentiate healthy people from people with PD, according to the "status" column of almost zero for good and 1 for PD. Information is in ASCII CSV format. CSV document lines include an example corresponding to a single voice recording. There are about six recordings of the affected person, the affected person's call is found in the first column.. The unique have a look at published characteristic extraction techniques for trendy voice problems.

MDVP:RAP	MDVP:PPQ	Jitter:DDP	MDVP:Shimmer	...	Shimmer:DDA	NHR	NHR	status	RPDE	DFA	spread1	spread2	D2	PPE
0.00370	0.00554	0.01109	0.04374	...	0.06545	0.02211	21.033	1	0.414783	0.815285	-4.813031	0.266482	2.301442	0.284654
0.00465	0.00696	0.01394	0.06134	...	0.09403	0.01929	19.085	1	0.458359	0.819521	-4.075192	0.335590	2.486855	0.368674
0.00544	0.00781	0.01633	0.05233	...	0.08270	0.01309	20.651	1	0.429895	0.825288	-4.443179	0.311173	2.342259	0.332634
0.00502	0.00698	0.01505	0.05492	...	0.08771	0.01353	20.644	1	0.434969	0.819235	-4.117501	0.334147	2.405554	0.368975
0.00655	0.00908	0.01966	0.06425	...	0.10470	0.01767	19.649	1	0.417356	0.823484	-3.747787	0.234513	2.332180	0.410335

Fig 3.3 Data Attributes

PPE	DFA	RPDE	numPulses	numPeriod	meanPeriod	stdDevPeriod	locPctJitter	locAbsJitter	rapJitter	ppq5Jitter	ddpJitter	locShimmer	locDbShim	apq3Shimr
0.85247	0.71826	0.57227	240	239	0.008064	8.68E-05	0.00218	1.76E-05	0.00067	0.00129	0.002	0.05883	0.517	0.03011
0.76686	0.69481	0.53966	234	233	0.008258	7.31E-05	0.00195	1.61E-05	0.00052	0.00112	0.00157	0.05516	0.502	0.0232
0.85083	0.67604	0.58982	232	231	0.00834	6.04E-05	0.00176	1.47E-05	0.00057	0.00111	0.00171	0.09902	0.897	0.05094
0.41121	0.79672	0.59257	178	177	0.010858	0.000183	0.00419	4.55E-05	0.00149	0.00268	0.00446	0.05451	0.527	0.02395
0.3279	0.79782	0.53028	236	235	0.008162	0.002669	0.00535	4.37E-05	0.00166	0.00227	0.00499	0.0561	0.497	0.02909
0.5078	0.78744	0.65451	226	221	0.007631	0.002696	0.00783	5.97E-05	0.00232	0.00312	0.00697	0.07752	0.678	0.03805
0.76095	0.62145	0.54543	322	321	0.005991	0.000107	0.00222	1.33E-05	0.00036	0.00094	0.00108	0.03203	0.28	0.0155
0.83671	0.62079	0.51179	318	317	0.006074	0.000136	0.00282	1.71E-05	0.00034	0.00088	0.00103	0.063	0.539	0.02949
0.80826	0.61766	0.50447	318	317	0.006057	6.93E-05	0.00161	9.73E-06	0.00027	0.00068	0.00081	0.02783	0.244	0.01376
0.85302	0.62247	0.54855	493	492	0.00391	3.99E-05	0.00075	2.93E-06	9.00E-05	0.00025	0.00027	0.0567	0.512	0.02692
0.80657	0.67256	0.61745	488	487	0.003956	5.38E-05	0.00083	3.29E-06	0.0001	0.00026	0.00029	0.06639	0.641	0.03747
0.82653	0.58326	0.44555	498	497	0.003873	3.26E-05	0.00069	2.68E-06	7.00E-05	0.00021	0.00022	0.02531	0.218	0.01283
0.8726	0.78996	0.78026	492	491	0.003924	6.72E-05	0.0028	1.10E-05	0.00077	0.00184	0.0023	0.20811	1.814	0.08936

Fig 3.4 Data Cleaning

Date Set Features

Types of Data Set

The data used in this study were gathered from 188 patients with PD (107 men and 81 women) with ages ranging from 33 to 87 (65.1 ± 10.9) at the Department of Neurology in Cerrahpaşaa Faculty of Medicine, Istanbul University. The control group consists of 64 healthy individuals (23 men and 41 women) with ages varying between 41 and 82 (61.1 ± 8.9). During the data collection process, the microphone is set to 44.1 KHz and following the physician's examination, the sustained phonation of the vowel /a/ was collected from each subject with three repetitions.

Number of Attributes, fields, description of the data set.

The Parkinson's disease database will help us to figure out whether the respective target is or is not having the disease; it's a multivariate data set. The database is the voice samples which have been accumulated from 31 people out of which 23 are having the disease. This data set is composed of the range of biomedical voice measurements and each column describes a particular voice measure and each row corresponds to one of the 195 voice recordings from these individuals. With the help of machine learning techniques, Machine Learning Techniques Will be creating a model which can be 100% accurate in the description of the patient. The model will analyze the data in the given data set and will protect whether the patient is or is not having Parkinson disease.

Qualification Information:

Matrix column entries (attributes):

- Name - ASCII title name and recording number
- MDVP: Fo (Hz) - Basic voice frequency
- MDVP: Fhi (Hz) - The frequency of the basic voice
- MDVP: Flo (Hz) - Basic voice frequency
- MDVP: Jitter (%), MDVP: Jitter (Abs), MDVP: RAP, MDVP: PPQ, Jitter: DDP

Several estimates of basic frequency variability

- MDVP: Shimmer, MDVP: Shimmer (dB), Shimmer: APQ3, Shimmer: APQ5,

MDVP: APQ,

- Shimmer: DDA - A few steps for size variation
- NHR, HNR - Two levels of sound measurement and tone components invoice
- Condition - The health condition of the subject (one) - Parkinson's (zero)-is healthy

3.1 DATA CLEANING:

Data cleaning is an important step in any machine learning project, including those related to Parkinson's disease. The goal of data cleaning is to ensure that the data is accurate, complete, and consistent, so that the machine learning algorithm can make accurate predictions.

Here are some steps that you can follow for data cleaning for Parkinson's disease using machine learning:

- Data Collection: The first step is to collect the data from reliable sources such as medical journals, patient reports, and clinical trials. You can also use publicly available datasets such as the Parkinson's Disease Data Set from the UCI Machine Learning Repository.

- **Data Integration:** Next, you will need to integrate the data from different sources into a single dataset. You should check the consistency of the data and identify any missing values, duplicate records, or errors in the data.
- **Data Transformation:** In this step, you will transform the data into a format that is suitable for machine learning algorithms. This may include scaling the data, encoding categorical variables, and removing irrelevant features.
- **Data Reduction:** If the dataset is very large, you may need to reduce the number of features by selecting only the most important ones. This can help to improve the accuracy of the machine learning algorithm and reduce computational costs.
- **Data Imputation:** If there are any missing values in the dataset, you will need to impute them using techniques such as mean imputation or regression imputation.
- **Data Normalization:** Normalize the dataset to scale all features within the same range, otherwise, some features might have a higher influence on the model than others.
- **Data Split:** Split the dataset into training and testing sets, in order to be able to test the performance of the model on unseen data.

These are some of the basic steps involved in data cleaning for Parkinson's disease using machine learning. It is important to ensure that the data is cleaned and prepared properly before using it to train machine learning models.

Importing Libraries import

pandas as pd import

numpy as np import os,sys

import xgboost as xgb

from sklearn.preprocessing import MinMaxScaler

from sklearn.model_selection import train_test_split

from sklearn.metrics import accuracy_score import

matplotlib.pyplot as plt import seaborn as sns import

warnings

warnings.filterwarnings('ignore')

Load Dataset

df=pd.read_csv("/content/Parkinson disease.csv") df

	name	MDVP:F0(Hz)	MDVP:F1(Hz)	MDVP:F2(Hz)	MDVP:jitter(%)	MDVP:jitter(Abs)	MDVP:RAP	MDVP:PPQ	Jitter:DDP	MDVP:Shimmer	...	Shimmer:DDA	NHR	HNR	status	RPDE	DFA	spread1	spread2	D2	PPE
0	phon_R01_S01_1	119.992	157.302	74.997	0.00784	0.00007	0.00370	0.00554	0.01109	0.04374	...	0.06545	0.02211	21.033	1	0.414783	0.815385	-4.813031	0.266482	2.301442	0.284654
1	phon_R01_S01_2	122.400	148.650	113.819	0.00968	0.00008	0.00465	0.00696	0.01394	0.06134	...	0.09403	0.01929	19.085	1	0.458359	0.819521	-4.075192	0.335590	2.486855	0.368674
2	phon_R01_S01_3	116.682	131.111	111.555	0.01050	0.00009	0.00544	0.00781	0.01633	0.05233	...	0.08270	0.01309	20.651	1	0.429895	0.825288	-4.443179	0.311173	2.342259	0.332634
3	phon_R01_S01_4	116.676	137.871	111.366	0.00997	0.00009	0.00502	0.00698	0.01505	0.05492	...	0.08771	0.01353	20.644	1	0.434969	0.819235	-4.117501	0.334147	2.405554	0.368975
4	phon_R01_S01_5	116.014	141.781	110.655	0.01284	0.00011	0.00655	0.00908	0.01966	0.06425	...	0.10470	0.01767	19.649	1	0.417356	0.823484	-3.747787	0.234513	2.332180	0.410335
...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...
190	phon_R01_S50_2	174.188	230.978	94.261	0.00459	0.00003	0.00263	0.00259	0.00790	0.04087	...	0.07008	0.02764	19.517	0	0.448439	0.657899	-6.538586	0.121952	2.657476	0.133050
191	phon_R01_S50_3	209.516	253.017	89.488	0.00564	0.00003	0.00331	0.00292	0.00994	0.02751	...	0.04812	0.01810	19.147	0	0.431674	0.683244	-6.195325	0.129303	2.784312	0.168895
192	phon_R01_S50_4	174.688	240.005	74.287	0.01360	0.00008	0.00624	0.00564	0.01873	0.02308	...	0.03804	0.10715	17.883	1	0.407567	0.655683	-6.787197	0.158453	2.679772	0.131728
193	phon_R01_S50_5	198.764	396.961	74.904	0.00740	0.00004	0.00370	0.00390	0.01109	0.02296	...	0.03794	0.07223	19.020	1	0.451221	0.643956	-6.744577	0.207454	2.138608	0.123306
194	phon_R01_S50_6	214.289	260.277	77.973	0.00567	0.00003	0.00295	0.00317	0.00885	0.01884	...	0.03078	0.04398	21.209	0	0.462803	0.664357	-5.724056	0.190667	2.555477	0.148569

195 rows x 24 columns

Fig. 3.5 Renaming columns

Renaming columns

```
df.rename(columns=({'MDVP:Fo(Hz)': 'avg_fre', 'MDVP:Fhi(Hz)': 'max_fre',
'MDVP:Flo(Hz)': 'min_fre', 'MDVP:Jitter(%)': 'var_fre1',
'MDVP:Jitter(Abs)': 'var_fre2',
'MDVP:RAP': 'var_fre3', 'MDVP:PPQ': 'var_fre4', 'Jitter:DDP': 'var_fre5',
'MDVP:Shimmer': 'var_amp1', 'MDVP:Shimmer(dB)': 'var_amp2',
'Shimmer:APQ3': 'var_amp3', 'Shimmer:APQ5': 'var_amp4', 'MDVP:APQ': 'var_amp5',
'Shimmer:DDA': 'var_amp6'}), inplace=True) df
```

	name	avg_fre	max_fre	min_fre	var_fre1	var_fre2	var_fre3	var_fre4	var_fre5	var_amp1	...	var_amp6	NHR	HNR	status	RPDE	DFA	spread1	spread2	D2	PPE
0	phon_R01_S01_1	119.992	157.302	74.997	0.00784	0.00007	0.00370	0.00554	0.01109	0.04374	...	0.06545	0.02211	21.033	1	0.414783	0.815285	-4.813031	0.266482	2.301442	0.284654
1	phon_R01_S01_2	122.400	148.650	113.819	0.00968	0.00008	0.00465	0.00696	0.01394	0.06134	...	0.09403	0.01929	19.085	1	0.458359	0.819521	-4.075192	0.335590	2.486855	0.368674
2	phon_R01_S01_3	116.682	131.111	111.555	0.01050	0.00009	0.00544	0.00781	0.01633	0.05233	...	0.08270	0.01309	20.651	1	0.429895	0.825288	-4.443179	0.311173	2.342259	0.332634
3	phon_R01_S01_4	116.676	137.871	111.366	0.00997	0.00009	0.00502	0.00698	0.01505	0.05492	...	0.08771	0.01353	20.644	1	0.434969	0.819235	-4.117501	0.334147	2.405554	0.368975
4	phon_R01_S01_5	116.014	141.781	110.655	0.01284	0.00011	0.00655	0.00908	0.01966	0.06425	...	0.10470	0.01767	19.649	1	0.417356	0.823484	-3.747787	0.234513	2.332180	0.410335
...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...
190	phon_R01_S50_2	174.188	230.978	94.261	0.00459	0.00003	0.00263	0.00259	0.00790	0.04087	...	0.07008	0.02764	19.517	0	0.448439	0.657899	-6.538586	0.121952	2.657476	0.133050
191	phon_R01_S50_3	209.516	253.017	89.488	0.00564	0.00003	0.00331	0.00292	0.00994	0.02751	...	0.04812	0.01810	19.147	0	0.431674	0.683244	-6.195325	0.129303	2.784312	0.168895
192	phon_R01_S50_4	174.688	240.005	74.287	0.01360	0.00008	0.00624	0.00564	0.01873	0.02308	...	0.03804	0.10715	17.883	1	0.407567	0.655683	-6.787197	0.158453	2.679772	0.131728
193	phon_R01_S50_5	198.764	396.961	74.904	0.00740	0.00004	0.00370	0.00390	0.01109	0.02296	...	0.03794	0.07223	19.020	1	0.451221	0.643956	-6.744577	0.207454	2.138608	0.123306
194	phon_R01_S50_6	214.289	260.277	77.973	0.00567	0.00003	0.00295	0.00317	0.00885	0.01884	...	0.03078	0.04398	21.209	0	0.462803	0.664357	-5.724056	0.190667	2.555477	0.148569

195 rows x 24 columns

Fig 3.6 Renamed columns

we will modify our data for proper and correct analysis so that we will check the unique values in the name and status column. After that we will remove the name variable from our data set because the name variable is not important for making predictions. After that, we see the shape of our data set when the value of status is one and 0. After seeing the shape we will see that whether our data is having missing values or not, if the data is having missing values, we will have to do appropriate calculations to remove it but, in our case, data didn't have any missing values therefore it was ready for further analysis.

Some screenshots of the process have been attached below:

<pre>[] len(df.name.unique())</pre> <pre>195</pre>	<pre>[] df.shape</pre> <pre>(195, 23)</pre>
<pre>[] len(df.status.unique())</pre> <pre>2</pre>	<pre>df[df['status']==1].shape</pre> <pre>(147, 23)</pre>
<pre>[] #we dont need the 'name' variable for predictions</pre> <pre>df = df.drop(['name'], axis = 1)</pre>	<pre>[] df[df['status']==0].shape</pre> <pre>(48, 23)</pre>
<pre>[] df.columns</pre>	

Fig 3.7 Processing of Dataset

Dimensions of Dataset df.shape

(195, 24)

Peak of the Data df.head(20)

	name	avg_fre	max_fre	min_fre	var_fre1	var_fre2	var_fre3	var_fre4	var_fre5	var_amp1	...	var_amp6	NHR	HNR	status	RPDE	DFA	spread1	spread2	D2	PPE
0	phon_R01_S01_1	119.992	157.302	74.997	0.00784	0.00007	0.00370	0.00554	0.01109	0.04374	...	0.06545	0.02211	21.033	1	0.414783	0.815285	-4.813031	0.266482	2.301442	0.284654
1	phon_R01_S01_2	122.400	148.650	113.819	0.00968	0.00008	0.00465	0.00696	0.01394	0.06134	...	0.09403	0.01929	19.085	1	0.458359	0.819521	-4.075192	0.335590	2.486855	0.368674
2	phon_R01_S01_3	116.682	131.111	111.555	0.01050	0.00009	0.00544	0.00781	0.01633	0.05233	...	0.08270	0.01309	20.651	1	0.429895	0.825288	-4.443179	0.311173	2.342259	0.332634
3	phon_R01_S01_4	116.676	137.871	111.366	0.00997	0.00009	0.00502	0.00698	0.01505	0.05492	...	0.08771	0.01353	20.644	1	0.434969	0.819235	-4.117501	0.334147	2.405554	0.368975
4	phon_R01_S01_5	116.014	141.781	110.655	0.01284	0.00011	0.00655	0.00908	0.01966	0.06425	...	0.10470	0.01767	19.649	1	0.417356	0.823484	-3.747787	0.234513	2.332180	0.410335
5	phon_R01_S01_6	120.552	131.162	113.787	0.00968	0.00008	0.00463	0.00750	0.01388	0.04701	...	0.06985	0.01222	21.378	1	0.415564	0.825069	-4.242867	0.299111	2.187560	0.357775
6	phon_R01_S02_1	120.267	137.244	114.820	0.00333	0.00003	0.00155	0.00202	0.00466	0.01608	...	0.02337	0.00607	24.886	1	0.596040	0.764112	-5.634322	0.257682	1.854785	0.211756
7	phon_R01_S02_2	107.332	113.840	104.315	0.00290	0.00003	0.00144	0.00182	0.00431	0.01567	...	0.02487	0.00344	26.892	1	0.637420	0.763262	-6.167603	0.183721	2.064693	0.163755
8	phon_R01_S02_3	95.730	132.068	91.754	0.00551	0.00006	0.00293	0.00332	0.00880	0.02093	...	0.03218	0.01070	21.812	1	0.615551	0.773587	-5.498678	0.327769	2.322511	0.231571
9	phon_R01_S02_4	95.056	120.103	91.226	0.00532	0.00006	0.00268	0.00332	0.00803	0.02838	...	0.04324	0.01022	21.862	1	0.547037	0.798463	-5.011879	0.325996	2.432792	0.271362
10	phon_R01_S02_5	88.333	112.240	84.072	0.00505	0.00006	0.00254	0.00330	0.00763	0.02143	...	0.03237	0.01166	21.118	1	0.611137	0.776156	-5.249770	0.391002	2.407313	0.249740
11	phon_R01_S02_6	91.904	115.871	86.292	0.00540	0.00006	0.00281	0.00336	0.00844	0.02752	...	0.04272	0.01141	21.414	1	0.583390	0.792520	-4.960234	0.363566	2.642476	0.275931
12	phon_R01_S04_1	136.926	159.866	131.276	0.00293	0.00002	0.00118	0.00153	0.00355	0.01259	...	0.01968	0.00581	25.703	1	0.460600	0.646486	-6.547148	0.152813	2.041277	0.138512
13	phon_R01_S04_2	139.173	179.139	76.556	0.00390	0.00003	0.00165	0.00208	0.00496	0.01642	...	0.02184	0.01041	24.889	1	0.430166	0.665833	-5.660217	0.254989	2.519422	0.199889
14	phon_R01_S04_3	152.845	163.305	75.836	0.00294	0.00002	0.00121	0.00149	0.00364	0.01828	...	0.03191	0.00609	24.922	1	0.474791	0.654027	-6.105098	0.203653	2.125618	0.170100
15	phon_R01_S04_4	142.167	217.455	83.159	0.00369	0.00003	0.00157	0.00203	0.00471	0.01503	...	0.02316	0.00839	25.175	1	0.565924	0.658245	-5.340115	0.210185	2.205546	0.234589
16	phon_R01_S04_5	144.188	349.259	82.764	0.00544	0.00004	0.00211	0.00292	0.00632	0.02047	...	0.02908	0.01859	22.333	1	0.567380	0.644692	-5.440040	0.239764	2.264501	0.218164
17	phon_R01_S04_6	168.778	232.181	75.603	0.00718	0.00004	0.00284	0.00387	0.00853	0.03327	...	0.04322	0.02919	20.376	1	0.631099	0.605417	-2.931070	0.434326	3.007463	0.430788
18	phon_R01_S05_1	153.046	175.829	68.623	0.00742	0.00005	0.00364	0.00432	0.01092	0.05517	...	0.07413	0.03160	17.280	1	0.665318	0.719467	-3.949079	0.357870	3.109010	0.377429
19	phon_R01_S05_2	156.405	189.398	142.822	0.00768	0.00005	0.00372	0.00399	0.01116	0.03995	...	0.05164	0.03365	17.153	1	0.649554	0.686080	-4.554466	0.340176	2.856676	0.322111

20 rows × 24 columns

Fig 3.8 Unwanted column

Statistical Summary df.describe()

	avg_fre	max_fre	min_fre	var_fre1	var_fre2	var_fre3	var_fre4	var_fre5	var_amp1	var_amp2	...	var_amp6	NHR	HNR	status	RPDE	DFA	spread1	spread2	D2	PPE
count	195.000000	195.000000	195.000000	195.000000	195.000000	195.000000	195.000000	195.000000	195.000000	195.000000	...	195.000000	195.000000	195.000000	195.000000	195.000000	195.000000	195.000000	195.000000	195.000000	195.000000
mean	154.228641	197.104918	116.324631	0.006220	0.000044	0.003306	0.003446	0.009920	0.029709	0.282251	...	0.046993	0.024847	21.885974	0.882051	0.498536	0.718099	-5.684397	0.226510	2.381826	0.206552
std	41.390065	91.491548	43.521413	0.004848	0.000035	0.002968	0.002759	0.008903	0.018857	0.194877	...	0.030459	0.040418	4.425764	0.323378	0.103942	0.055336	1.090208	0.083406	0.382799	0.090119
min	88.333000	102.145000	65.476000	0.001680	0.000007	0.000680	0.000920	0.002040	0.009540	0.085000	...	0.013640	0.000650	8.441000	0.000000	0.256570	0.574282	-7.964984	0.006274	1.423287	0.044539
25%	117.572000	134.862500	84.291000	0.003460	0.000020	0.001660	0.001860	0.004985	0.016505	0.148500	...	0.024735	0.005925	19.198000	1.000000	0.421306	0.674758	-6.450096	0.174351	2.099125	0.137451
50%	148.790000	175.829000	104.315000	0.004940	0.000030	0.002500	0.002690	0.007490	0.022970	0.221000	...	0.038360	0.011660	22.085000	1.000000	0.495954	0.722254	-5.720868	0.218885	2.361532	0.194052
75%	182.769000	224.205500	140.018500	0.007365	0.000060	0.003835	0.003955	0.011505	0.037885	0.350000	...	0.060795	0.025640	25.075500	1.000000	0.587562	0.761881	-5.046192	0.279234	2.636456	0.252980
max	260.105000	592.030000	239.170000	0.033160	0.000260	0.021440	0.019580	0.064330	0.119080	1.302000	...	0.169420	0.314820	33.047000	1.000000	0.685151	0.825288	-2.434031	0.450493	3.671155	0.527367

8 rows × 23 columns

Fig 3.9 Information of Data set


```

<class 'pandas.core.frame.DataFrame'>
RangeIndex: 195 entries, 0 to 194
Data columns (total 24 columns):
#   Column      Non-Null Count  Dtype
---  -
0   name        195 non-null    object
1   avg_fre     195 non-null    float64
2   max_fre     195 non-null    float64
3   min_fre     195 non-null    float64
4   var_fre1    195 non-null    float64
5   var_fre2    195 non-null    float64
6   var_fre3    195 non-null    float64
7   var_fre4    195 non-null    float64
8   var_fre5    195 non-null    float64
9   var_amp1    195 non-null    float64
10  var_amp2    195 non-null    float64
11  var_amp3    195 non-null    float64
12  var_amp4    195 non-null    float64
13  var_amp5    195 non-null    float64
14  var_amp6    195 non-null    float64
15  NHR         195 non-null    float64
16  HNR         195 non-null    float64
17  status      195 non-null    int64
18  RPDE        195 non-null    float64
19  DFA         195 non-null    float64
20  spread1     195 non-null    float64
21  spread2     195 non-null    float64
22  D2          195 non-null    float64
23  PPE         195 non-null    float64
dtypes: float64(22), int64(1), object(1)
memory usage: 36.7+ KB

```

Fig 3.10 Duplication of Data set

Duplicate Entries `df.duplicated().sum()`

0

unwanted columns

`df.drop(columns="name",axis=1,inplace=True)`

`df`

	avg_fre	max_fre	min_fre	var_fre1	var_fre2	var_fre3	var_fre4	var_fre5	var_amp1	var_amp2	...	var_amp6	NHR	HNR	status	RPDE	DFA	spread1	spread2	D2	PPE
0	119.992	157.302	74.997	0.00784	0.00007	0.00370	0.00554	0.01109	0.04374	0.426	...	0.06545	0.02211	21.033	1	0.414783	0.815285	-4.813031	0.266482	2.301442	0.284654
1	122.400	148.650	113.819	0.00968	0.00008	0.00465	0.00696	0.01394	0.06134	0.626	...	0.09403	0.01929	19.085	1	0.458359	0.819521	-4.075192	0.335590	2.486855	0.368674
2	116.682	131.111	111.555	0.01050	0.00009	0.00544	0.00781	0.01633	0.05233	0.482	...	0.08270	0.01309	20.651	1	0.429895	0.825288	-4.443179	0.311173	2.342259	0.332634
3	116.676	137.871	111.366	0.00997	0.00009	0.00502	0.00698	0.01505	0.05492	0.517	...	0.08771	0.01353	20.644	1	0.434969	0.819235	-4.117501	0.334147	2.405554	0.368975
4	116.014	141.781	110.655	0.01284	0.00011	0.00655	0.00908	0.01966	0.06425	0.584	...	0.10470	0.01767	19.649	1	0.417356	0.823484	-3.747787	0.234513	2.332180	0.410335
...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...
190	174.188	230.978	94.261	0.00459	0.00003	0.00263	0.00259	0.00790	0.04087	0.405	...	0.07008	0.02764	19.517	0	0.448439	0.657899	-6.538586	0.121952	2.657476	0.133050
191	209.516	253.017	89.488	0.00564	0.00003	0.00331	0.00292	0.00994	0.02751	0.263	...	0.04812	0.01810	19.147	0	0.431674	0.683244	-6.195325	0.129303	2.784312	0.168895
192	174.688	240.005	74.287	0.01360	0.00008	0.00624	0.00564	0.01873	0.02308	0.256	...	0.03804	0.10715	17.883	1	0.407567	0.655683	-6.787197	0.158453	2.679772	0.131728
193	198.764	396.961	74.904	0.00740	0.00004	0.00370	0.00390	0.01109	0.02296	0.241	...	0.03794	0.07223	19.020	1	0.451221	0.643956	-6.744577	0.207454	2.138608	0.123306
194	214.289	260.277	77.973	0.00567	0.00003	0.00295	0.00317	0.00885	0.01884	0.190	...	0.03078	0.04398	21.209	0	0.462803	0.664357	-5.724056	0.190667	2.555477	0.148569

195 rows × 23 columns

Fig 3.11 Handling of missing values

3.2 HANDLING MISSING VALUES:

Dealing with missing values in a dataset is an important task in machine learning as they can affect the performance and accuracy of a model. There are several techniques to handle missing values in a dataset, such as:

- **Removal of missing values:** One approach is to remove the rows or columns containing missing values. However, this approach can lead to loss of important information and reduction in the size of the dataset.
- **Imputation of missing values:** Another approach is to impute the missing values with an estimated value. There are several methods for imputation, such as mean imputation, median imputation, mode imputation, and regression imputation.

In the case of Parkinson's disease detection, missing values can be handled using the following steps:

- **Identify missing values:** The first step is to identify the missing values in the dataset. This can be done using various methods, such as visualization, summary statistics, or code.
- **Decide on the imputation method:** Depending on the amount and type of missing values, choose an appropriate imputation method. For instance, if there are only a few

missing values in a column, mean imputation can be used. If there are many missing values, regression imputation may be a better choice.

- Implement the imputation method: Once the imputation method is chosen, implement it in the dataset. This can be done using libraries such as pandas or scikit-learn in Python. □ Validate the imputation: After imputing the missing values, validate the imputed dataset using visualization or summary statistics. This step is crucial to ensure that the imputed values are not biasing the dataset.
- Train and evaluate the model: Finally, use the imputed dataset to train and evaluate the model. The accuracy and performance of the model should be compared with and without the imputed values.

Missing values

```
df.isnull().sum()
```

```
avg_fre      0
max_fre      0
min_fre      0
var_fre1     0
var_fre2     0
var_fre3     0
var_fre4     0
var_fre5     0
var_amp1     0
var_amp2     0
var_amp3     0
var_amp4     0
var_amp5     0
var_amp6     0
NHR          0
HNR          0
status       0
RPDE         0
DFA          0
spread1      0
spread2      0
D2           0
PPE          0
dtype: int64
```

Fig 3.12 Missing values

df.notnull()

	avg_fre	max_fre	min_fre	var_fre1	var_fre2	var_fre3	var_fre4	var_fre5	var_amp1	var_amp2	...	var_amp6	NHR	HNR	status	RPDE	DFA	spread1	spread2	D2	PPE
0	True	True	True	True	True	True	True	True	True	True	...	True	True	True	True	True	True	True	True	True	True
1	True	True	True	True	True	True	True	True	True	True	...	True	True	True	True	True	True	True	True	True	True
2	True	True	True	True	True	True	True	True	True	True	...	True	True	True	True	True	True	True	True	True	True
3	True	True	True	True	True	True	True	True	True	True	...	True	True	True	True	True	True	True	True	True	True
4	True	True	True	True	True	True	True	True	True	True	...	True	True	True	True	True	True	True	True	True	True
...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...
190	True	True	True	True	True	True	True	True	True	True	...	True	True	True	True	True	True	True	True	True	True
191	True	True	True	True	True	True	True	True	True	True	...	True	True	True	True	True	True	True	True	True	True
192	True	True	True	True	True	True	True	True	True	True	...	True	True	True	True	True	True	True	True	True	True
193	True	True	True	True	True	True	True	True	True	True	...	True	True	True	True	True	True	True	True	True	True
194	True	True	True	True	True	True	True	True	True	True	...	True	True	True	True	True	True	True	True	True	True

195 rows x 23 columns

Fig 3.13 Notnull values

df.columns

```
Index(['avg_fre', 'max_fre', 'min_fre', 'var_fre1', 'var_fre2', 'var_fre3',
      'var_fre4', 'var_fre5', 'var_amp1', 'var_amp2', 'var_amp3', 'var_amp4',
      'var_amp5', 'var_amp6', 'NHR', 'HNR', 'status', 'RPDE', 'DFA',
      'spread1', 'spread2', 'D2', 'PPE'],
      dtype='object')
```

df.skew()

```
avg_fre      0.591737
max_fre      2.542146
min_fre      1.217350
var_fre1     3.084946
var_fre2     2.649071
var_fre3     3.360708
var_fre4     3.073892
var_fre5     3.362058
var_amp1     1.666480
var_amp2     1.999389
var_amp3     1.580576
var_amp4     1.798697
var_amp5     2.618047
var_amp6     1.580618
NHR          4.220709
HNR         -0.514317
status      -2.387366
RPDE        -0.143402
DFA         -0.033214
spread1      0.432139
spread2      0.144430
D2           0.430384
PPE          0.797491
dtype: float64
```

Fig 3.14 Skewing of Data set

```
sns.countplot(data=df,x="status")
```

```
<Axes: xlabel='status', ylabel='count'>
```

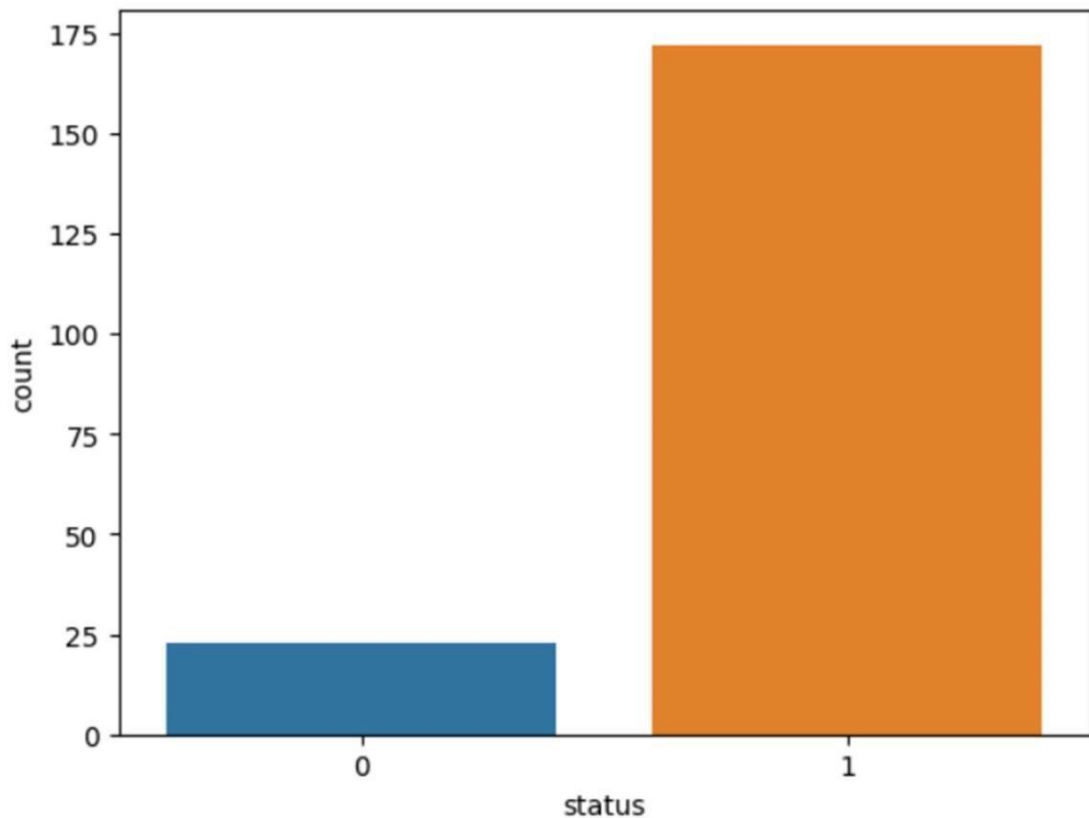


Fig 3.15 Countplot

3.3 FEATURE ENGINEERING:

Parkinson's disease is a neurodegenerative disorder that affects the nervous system. Machine learning algorithms can be used to detect Parkinson's disease by analyzing data from various sources, such as voice recordings, accelerometer data, and handwriting.

Feature engineering is the process of selecting and extracting the most relevant features from the data that can be used to train machine learning models. In the context of Parkinson's disease detection, feature engineering involves selecting the most informative features from the data sources and transforming them into a format that can be used by machine learning algorithms.

Here are some potential features that could be used in a Parkinson detection system:

- **Voice features:** People with Parkinson's disease often exhibit changes in their voice, such as tremors or reduced loudness. Voice features that could be used in a detection system include pitch, intensity, jitter, and shimmer.
- **Accelerometer features:** Accelerometers can be used to measure movement patterns in people with Parkinson's disease. Features that could be extracted from accelerometer data include the magnitude and frequency of movements, as well as the variability and regularity of those movements.
- **Handwriting features:** Parkinson's disease can affect a person's handwriting, resulting in changes in the size, shape, and speed of their writing. Handwriting features that could be used in a detection system include stroke length, duration, and pressure.
- **Clinical features:** In addition to data from sensors, clinical data such as age, gender, and medication history can also be used as features in a Parkinson detection system.

Once these features have been selected, they can be used to train machine learning models such as support vector machines, random forests, or neural networks. The models can then be evaluated on a validation dataset to determine their accuracy in detecting Parkinson's disease.

3.4 HANDLING CATEGORICAL VARIABLES

When building a Parkinson's disease detection system using machine learning, handling categorical variables is an important consideration. Categorical variables are variables that can take on a limited number of values, often representing a particular category or group.

In order to handle categorical variables, there are a few approaches that can be taken:

- **One-hot encoding:** This involves creating a new binary variable for each category in the original variable. For example, if we have a categorical variable "gender" with two categories "male" and "female", we can create two new binary variables "is_male" and "is_female". If a person is male, the "is_male" variable would be set to 1 and the

"is_female" variable would be set to 0. This approach can be effective, but can also lead to high-dimensional data and potential issues with overfitting.

- Target encoding: This involves replacing each category with the mean of the target variable for that category. For example, if we have a categorical variable "occupation" and we are trying to predict Parkinson's disease, we could replace each category with the mean Parkinson's disease status for individuals in that occupation. This approach can be effective, but can also be sensitive to outliers and may not work well for categories with very few observations.

Ultimately, the choice of how to handle categorical variables will depend on the specific data and problem at hand. It is important to carefully consider the potential advantages and disadvantages of each approach and to test multiple methods to determine which is most effective.

3.5 FEATURE SCALING

Feature scaling is one of the most important preprocessing techniques used in Machine Learning. It is the process of transforming numerical data into a common scale without affecting the actual meaning or importance of the data values. In machine learning, datasets often contain features with different ranges and units. For example, one feature may contain values between 0 and 1, while another feature may contain values in thousands. If these features are directly provided to a machine learning model, the algorithm may give more importance to features with larger values and ignore features with smaller values. This can reduce the performance and accuracy of the model. Therefore, feature scaling is performed to ensure that all features contribute equally during model training and prediction.

In the context of Parkinson's disease detection systems, feature scaling plays a major role because the dataset contains different types of medical and biological measurements such as voice recordings, accelerometer readings, handwriting features, and clinical data. These features naturally have different scales and ranges. By applying feature scaling, the values of all features are brought to a similar range, which helps machine learning algorithms learn patterns more effectively and make more accurate predictions.

One commonly used feature scaling technique is **Min-Max Scaling**, also known as normalization. This technique transforms the feature values into a fixed range, usually between 0 and 1. The minimum value of the feature becomes 0, and the maximum value becomes 1. All other values are adjusted proportionally within this range. Min-Max scaling is especially useful when the data does not follow a normal distribution and when preserving the relationship between data points is important.

For example, in Parkinson's disease detection, voice-related features such as pitch, intensity, jitter, shimmer, and frequency variations are commonly used for prediction. These voice features may have different numerical ranges. Pitch values may be high, while jitter values may be very small. If scaling is not applied, the algorithm may focus more on larger numerical features and neglect smaller but equally important features. By using Min-Max scaling, all voice features are normalized into a common range, ensuring balanced contribution during model training.

Another widely used feature scaling method is **Standardization**, also called Z-score normalization. This technique transforms the data so that the feature values have a mean of 0 and a standard deviation of 1. Standardization is useful when the data follows a normal distribution or when machine learning algorithms such as Support Vector Machines (SVM), Logistic Regression, and K-Nearest Neighbors (KNN) are used. The process involves subtracting the mean value of the feature from each data point and then dividing it by the standard deviation. As a result, the transformed data becomes centered around zero with uniform variance.

In Parkinson's disease prediction systems, accelerometer-based features such as movement magnitude, frequency, speed, and variability are often scaled using standardization. Accelerometer data collected from wearable devices or motion sensors can vary greatly in magnitude and distribution. Standardization helps in reducing these variations and ensures that the model treats all movement-related features equally. Similarly, handwriting-related features such as stroke length, writing pressure, speed, and duration are also standardized because these values can differ significantly between individuals. By scaling handwriting data, the system becomes more reliable in identifying patterns associated with Parkinson's disease.

Clinical features such as age, medication history, disease duration, and health measurements can also have different ranges and units. For instance, age may range from 40 to 90 years, while some medical scores may range only between 1 and 10. Applying Min-Max scaling or standardization to clinical data helps maintain consistency across the dataset and improves model performance.

Feature scaling not only improves the accuracy of machine learning models but also speeds up the training process. Many machine learning algorithms calculate distances, gradients, or statistical relationships between features. If the feature ranges are very different, the optimization process becomes slower and less efficient. Scaling helps the algorithms converge faster and improves stability during training. It also reduces the risk of bias toward features with larger numerical values.

After feature scaling is completed, the dataset is generally divided into three parts: training data, validation data, and testing data. The training dataset is used to train the machine learning model, the validation dataset is used to tune model parameters and prevent overfitting, and the testing dataset is used to evaluate the final performance of the model. Since all features are scaled uniformly, the machine learning model can better understand the relationships between features and produce more accurate predictions.

CHAPTER 4

EXPLORATORY DATA ANALYSIS

Exploratory Data Analysis (EDA) is a crucial step in building a Parkinson detection system using machine learning. EDA involves examining the dataset to gain insights into the data and identify any patterns or relationships that may exist between the features.

Here are some steps that can be followed in EDA for a Parkinson detection system:

1. **Data cleaning:** The first step in EDA is to clean the dataset by removing any missing or incorrect values. This can be done using techniques such as imputation or removal of missing values.
2. **Descriptive statistics:** Descriptive statistics such as mean, median, standard deviation, and range can be calculated for each feature to get an overall understanding of the data.
3. **Visualization:** Data visualization techniques such as histograms, box plots, and scatter plots can be used to identify any patterns or relationships that may exist between the features. For example, scatter plots can be used to visualize the relationship between two features, while box plots can be used to identify any outliers in the data.
4. **Correlation analysis:** Correlation analysis can be used to identify any linear relationships that may exist between the features. This can be done using techniques such as Pearson's correlation coefficient or Spearman's rank correlation coefficient.
5. **Feature selection:** Feature selection involves identifying the most relevant features that contribute to the prediction of Parkinson's disease. This can be done using techniques such as correlation analysis or feature importance scores from machine learning models.
6. **Class balance analysis:** It is important to ensure that the dataset has a balanced distribution of Parkinson's disease and non-Parkinson's disease cases. Class balance analysis can be done using techniques such as frequency counts or visualization techniques such as bar plots.

The insights gained from EDA can help in selecting the appropriate machine learning models and optimizing their hyperparameters. EDA can also help in identifying any limitations or biases in the dataset that may affect the accuracy of the models.

System Design

The Proposed Architectures has totally 4 parts where it involves Dataset i.e., Data Acquisition, Feature Extraction, Classification and Output production as shown below in figure.

- Data Acquisition involves acquiring all the data available which involves voice samples of People which contains noise or noiseless Features.
- Feature Extraction involves Voice analysis based on its Features i.e. MDVP.Jitter etc.
- Classification involves processing the given features of dataset using different classifiers such as SVM, XGBoost and Classification, Regression using Random Tree Classifier and Logistic Regression.
- Final Output is Produced for Prediction of the Disease involving individual Interests.

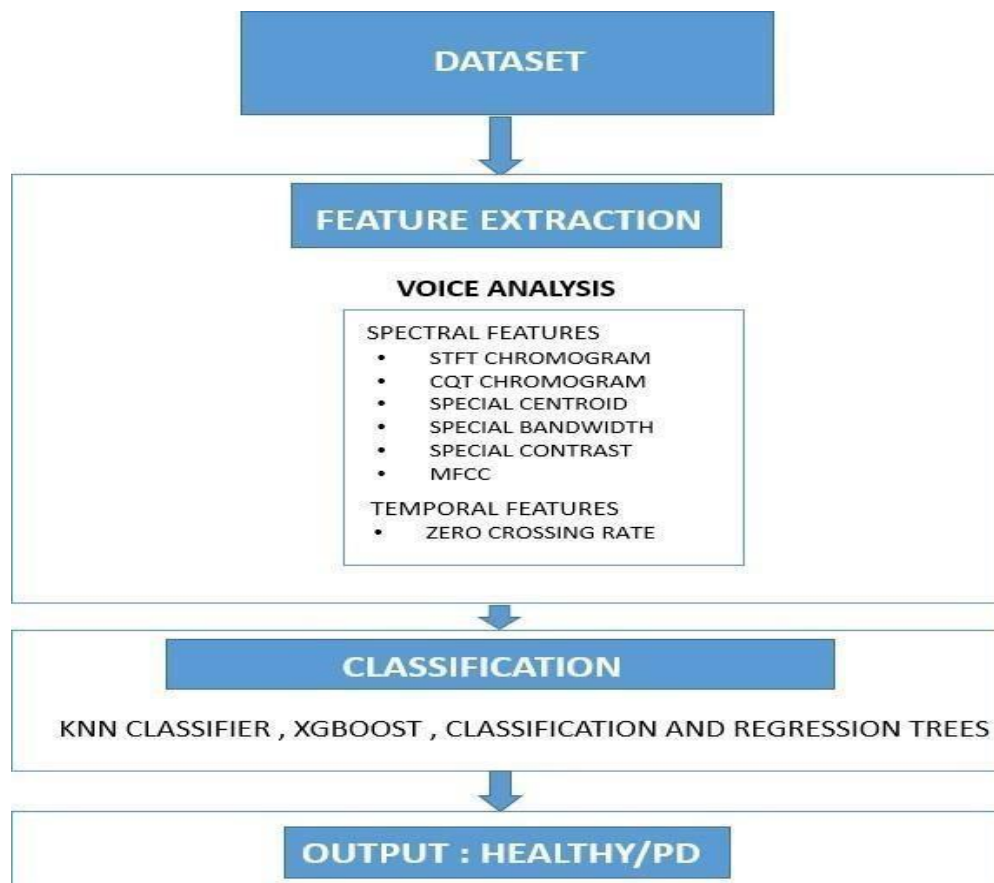


Fig 4.1 : System Design

Dataflow Diagram :

The Data flow diagram explains the basic flow data through various steps of Parkinson's disease Detection.

- In the first step we will collect data like from the patient with different medical equipment.
- The collected data is now sent process for training the data which will be classified using different ML algorithms.
- After classification the data into Train and Test datasets, the datasets will be sent to predict whether the patient has the disease or not.

The below figure3 shows the dataflow diagram for the proposed solution for the detection of Parkinson's Disease.

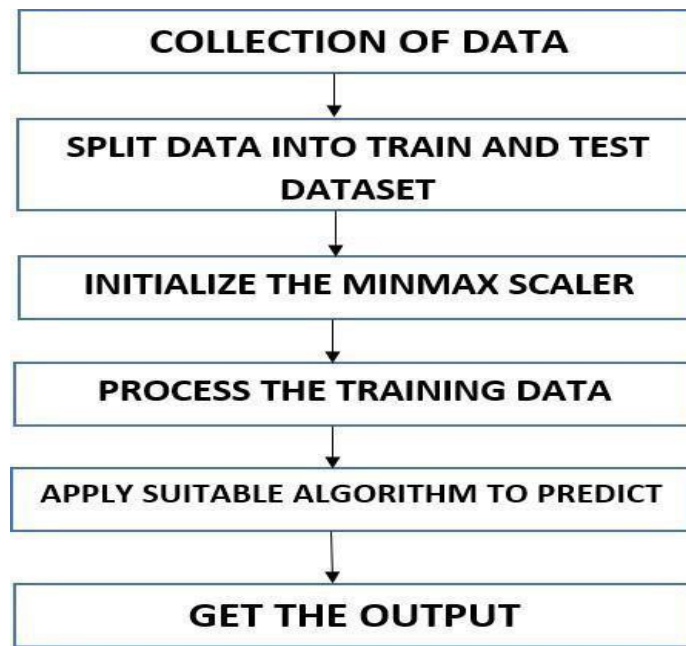


Fig 4.2 : Dataflow Diagram to detect

The Figure gives the combined analysis of both voice and image analysis to detect Parkinson's disease.

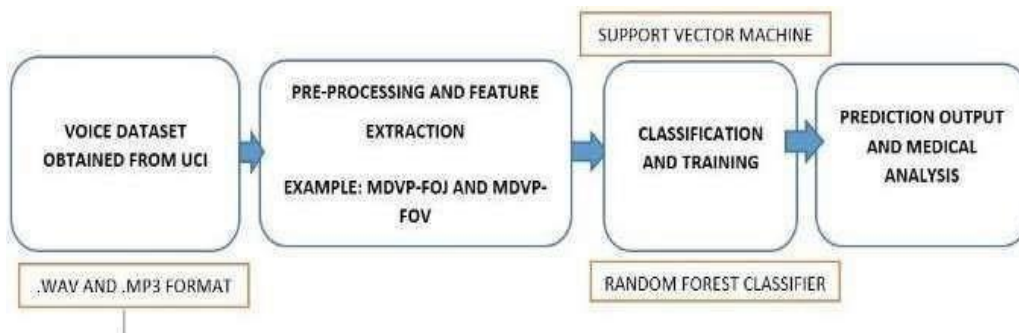


Fig 4.3 : Data flow diagram of voice

Class Diagram

The Fig explains the class diagram of Parkinson's, detection.

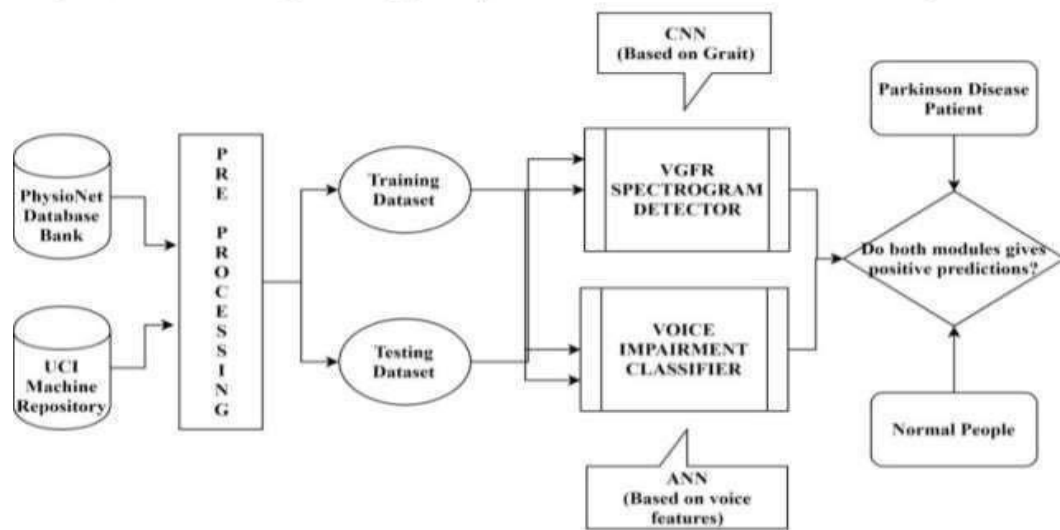


Fig 4.4 Class diagram

4.1 DESCRIPTIVE STATISTICS:

Descriptive statistics are used to summarize and describe the key characteristics of a dataset. In the context of a Parkinson detection system using machine learning, some common descriptive statistics that can be calculated include:

- **Mean:** The average value of a feature or input variable can provide insight into its central tendency. For example, the mean value of a patient's tremor intensity over a given time period can help identify the severity of their Parkinson's symptoms.
- **Standard Deviation:** The standard deviation measures the amount of variation or spread in a dataset. It can be used to identify outliers or to determine the level of precision of a measurement. For instance, the standard deviation of a patient's voice pitch can indicate how much variability exists in their speech patterns.
- **Median:** The median value represents the middle point of a dataset, where half the values are above and half are below. It can be used to describe the distribution of a feature when there are extreme values or outliers in the data. For example, the median value of a patient's gait speed can provide a more accurate representation of their walking ability than the mean, if some values are abnormally high or low.

- **Range:** The range represents the difference between the maximum and minimum values in a dataset. It can provide a quick summary of the overall spread of the data. For instance, the range of a patient's reaction time can indicate the level of cognitive impairment associated with their Parkinson's disease.
- **Correlation:** Correlation measures the degree of association between two variables. It can help identify relationships between features that may be relevant for Parkinson's diagnosis or monitoring. For example, the correlation between a patient's age and their disease onset may provide insight into the etiology of Parkinson's disease.

These are just a few examples of the many descriptive statistics that can be used to analyze data in a Parkinson detection system using machine learning. The choice of which statistics to use will depend on the specific research question, the type of data being analyzed, and the goals of the analysis.

4.2 DISTRIBUTION OF THE TARGET VARIABLE

- The distribution of the target variable in a Parkinson Detection System using Machine Learning would depend on the specific dataset being used and the classification task being performed.
- In general, the target variable in a Parkinson Detection System is a binary variable that indicates whether a patient has Parkinson's disease (PD) or not. Therefore, the distribution of the target variable would be a distribution of binary labels, with one class indicating the presence of PD and the other class indicating the absence of PD.
- The distribution of these two classes would depend on the population from which the data was collected. For example, if the dataset was collected from a population with a high prevalence of PD, then the distribution of the target variable may be skewed towards the positive class (PD present). Conversely, if the dataset was collected from a population with a low prevalence of PD, then the distribution of the target variable may be skewed towards the negative class (PD absent).
- It's also possible that the distribution of the target variable is balanced, with an equal number of instances in each class. This could be achieved through careful selection of the data or through data augmentation techniques.

- Ultimately, the distribution of the target variable in a Parkinson Detection System using Machine Learning will depend on the specific dataset and classification task being performed, and may vary widely from one study to another.

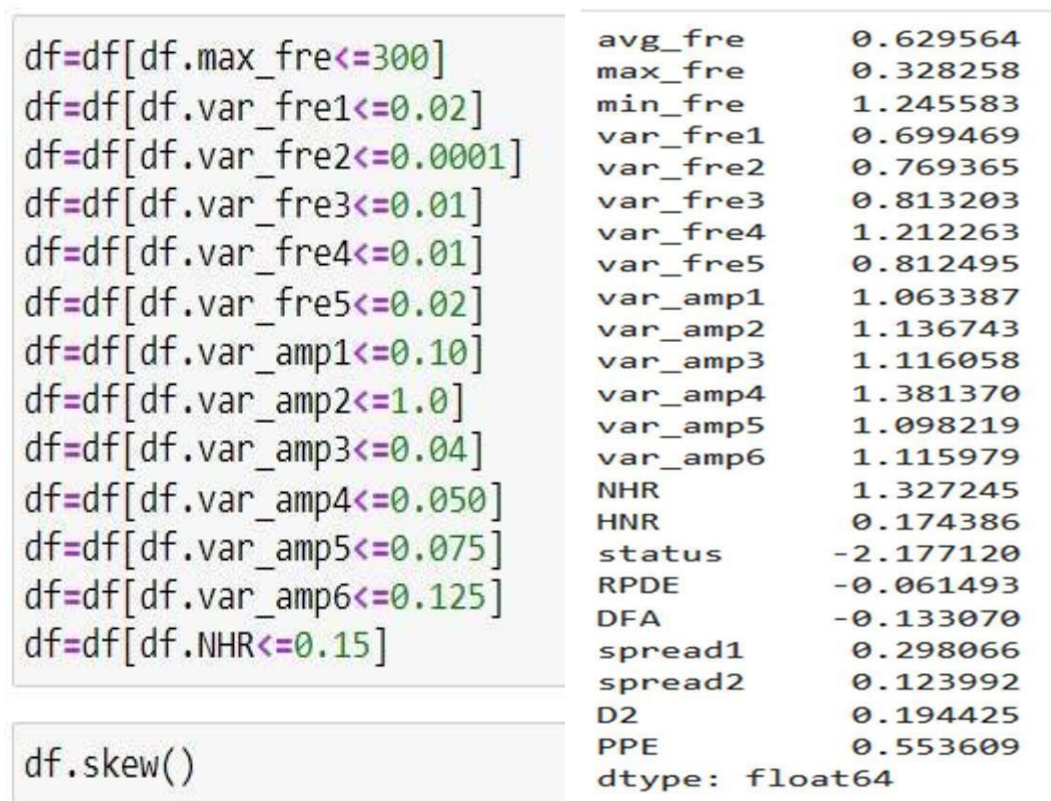


Fig 4.5 Skewing


```
sns.distplot(df['NHR'])  
<Axes: xlabel='NHR', ylabel='Density'>
```

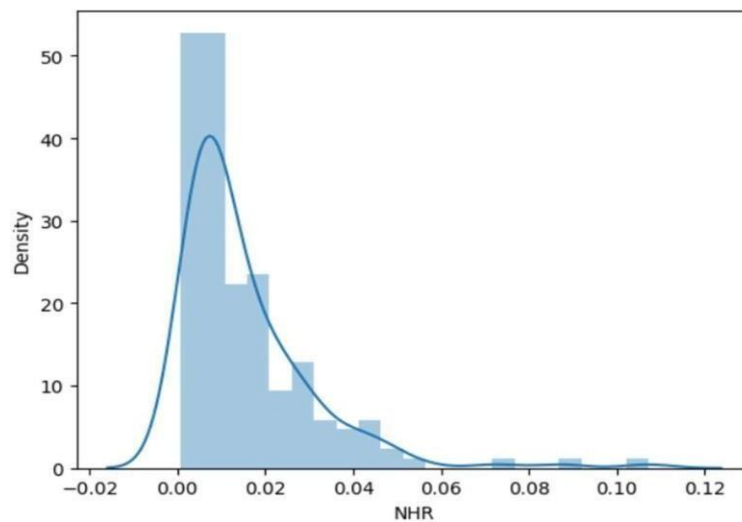


Fig 4.6 Label of Density

4.3 CORRELATION ANALYSIS

- Parkinson's disease is a progressive neurological disorder that affects movement and speech. Early detection and diagnosis of Parkinson's disease can help patients receive timely treatment and management of their symptoms. Machine learning has emerged as a promising tool for the early detection and diagnosis of Parkinson's disease.
- Correlation analysis is a statistical technique used to determine the strength and direction of the relationship between two variables. In the context of a Parkinson detection system using machine learning, correlation analysis can be used to determine the relationship between various features (such as tremor amplitude, gait speed, etc.) and the likelihood of Parkinson's disease.
- One common approach for using machine learning in Parkinson detection systems is to use supervised learning algorithms. In supervised learning, the machine learning model is trained on a labeled dataset, where each data point is associated with a label indicating whether the individual has Parkinson's disease or not.

- After training the model, the correlation between the different features and the predicted label can be analyzed to determine which features are most strongly associated with Parkinson's disease. This can help identify the most important features to include in a Parkinson detection system.
- In addition to correlation analysis, other statistical techniques such as logistic regression and decision tree analysis can also be used to analyze the relationship between features and Parkinson's disease. Ultimately, the goal of these analyses is to develop an accurate and reliable Parkinson detection system that can aid in the early diagnosis and management of the disease.

A correlation matrix is a table showing correlation coefficients between variables. Each cell in the table shows the correlation between two variables. A correlation matrix is used to summarize data, as an input into a more advanced analysis, and as a diagnostic for advanced analyses.

```
correl=x1.corr() plt.figure(figsize=(20,20)) sns.heatmap(correl,annot=True,cmap='OrRd')  
plt.show()
```

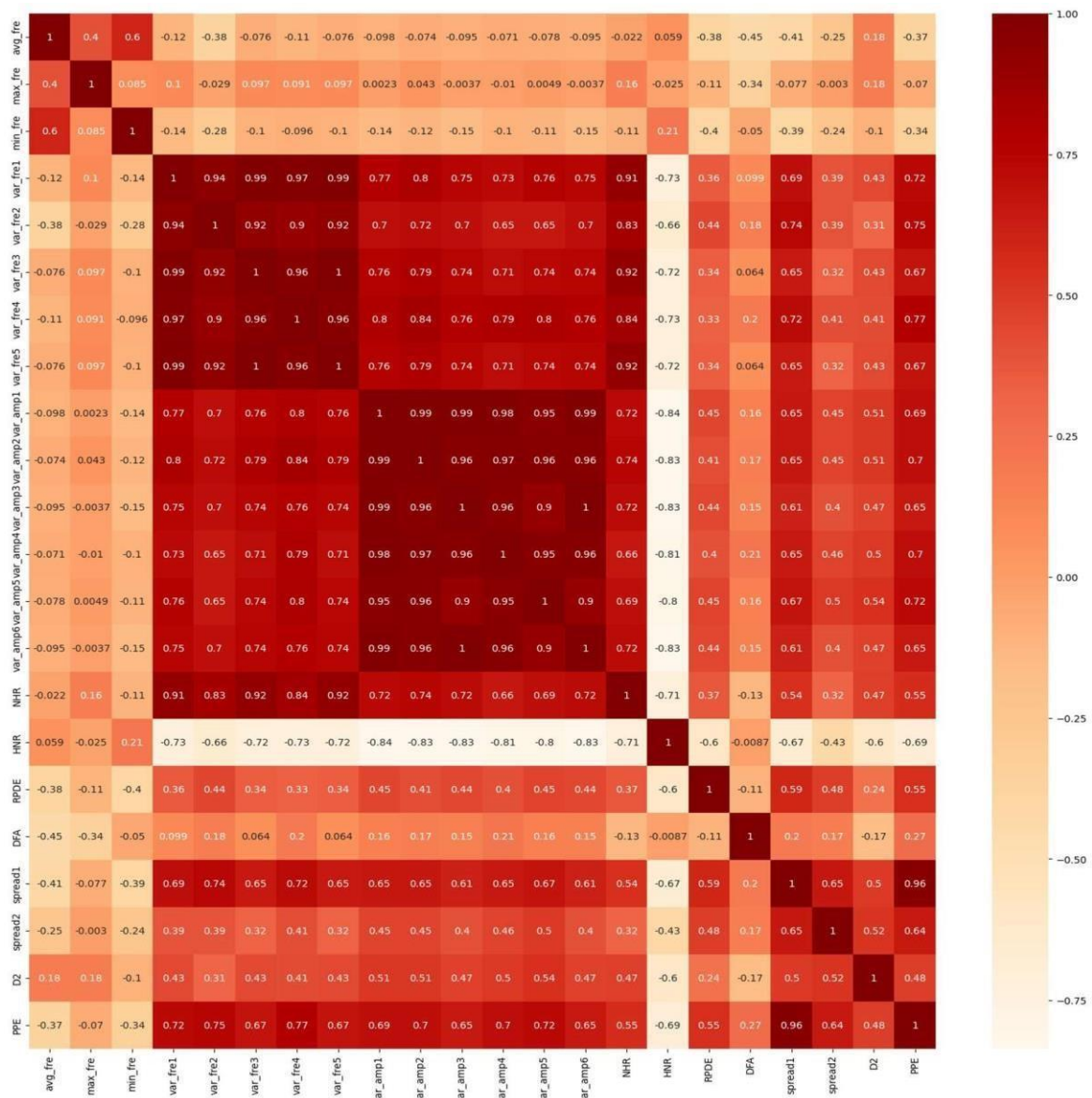
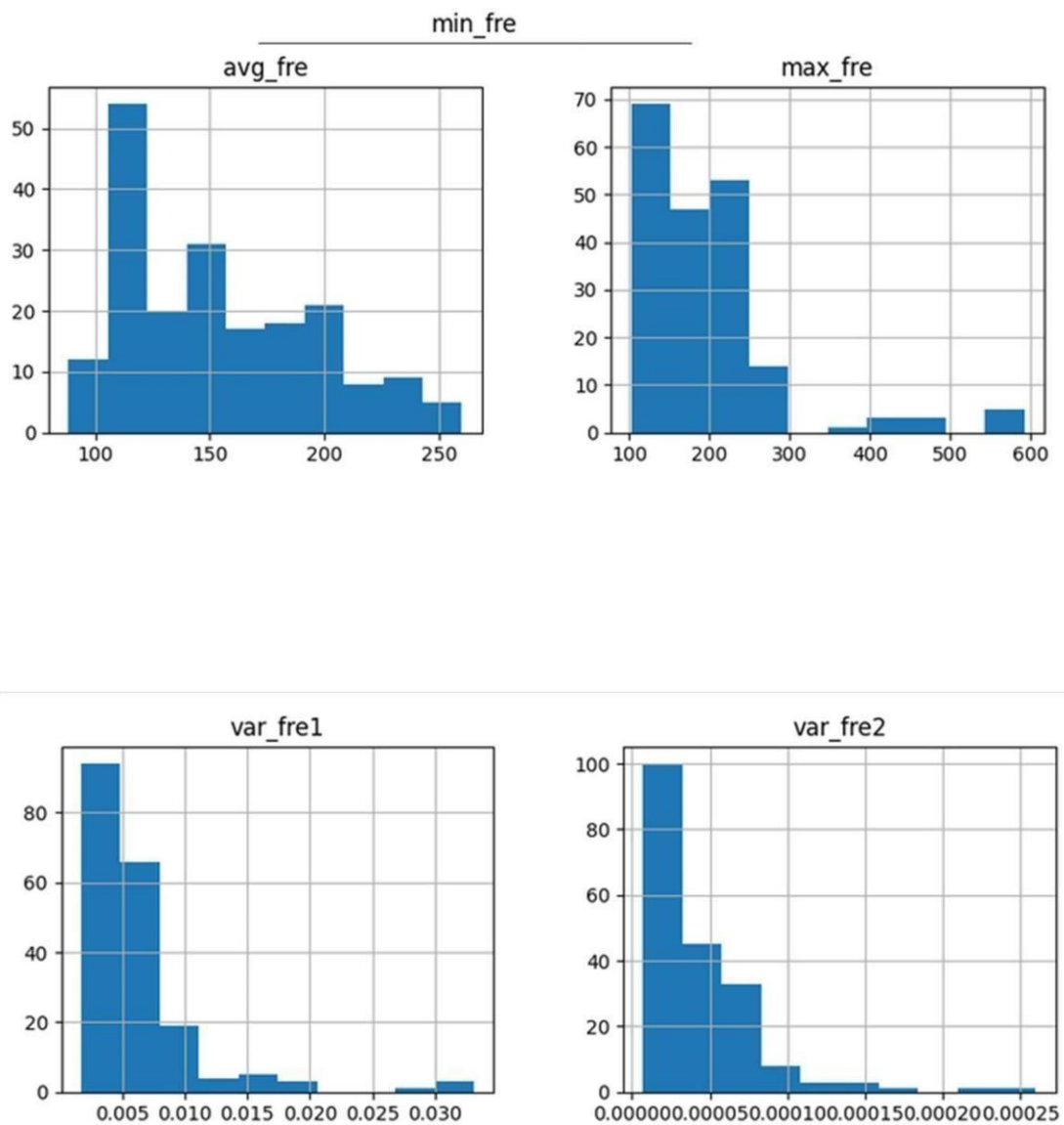
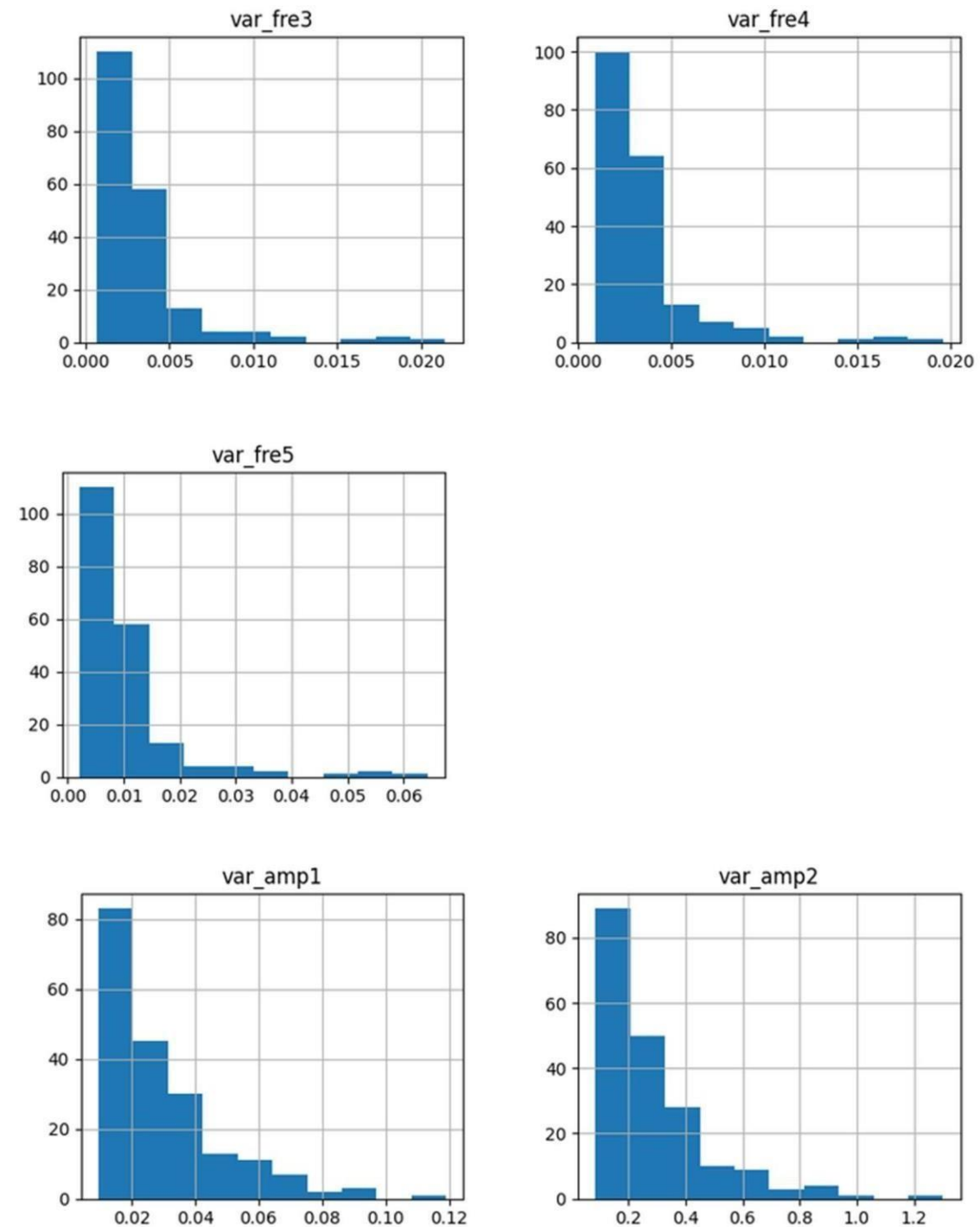
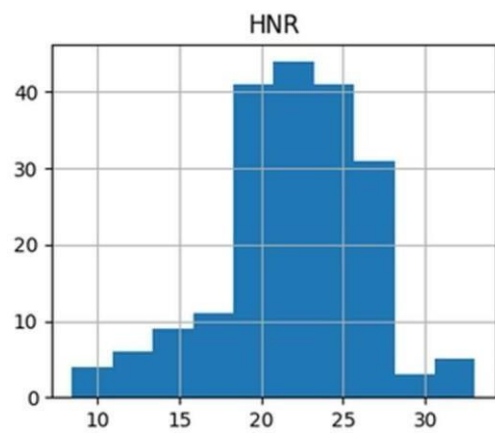
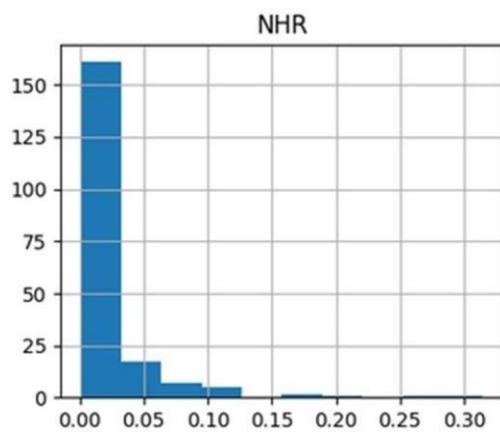
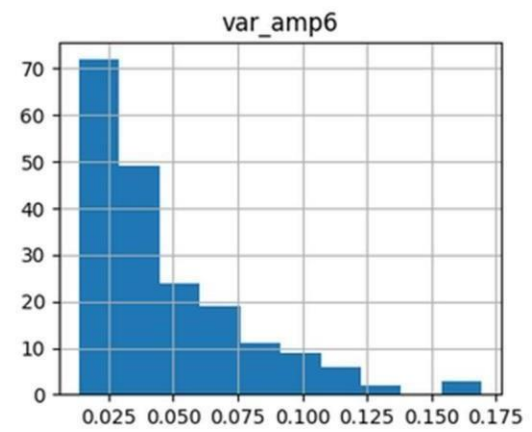
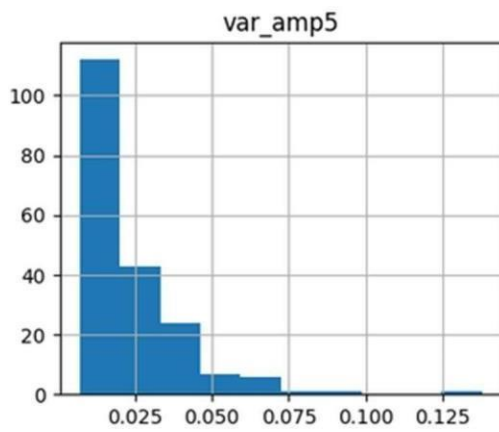
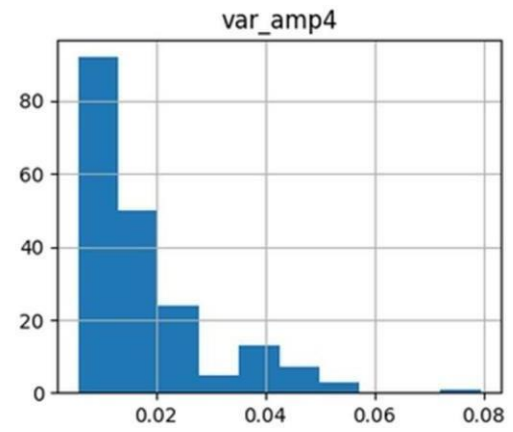
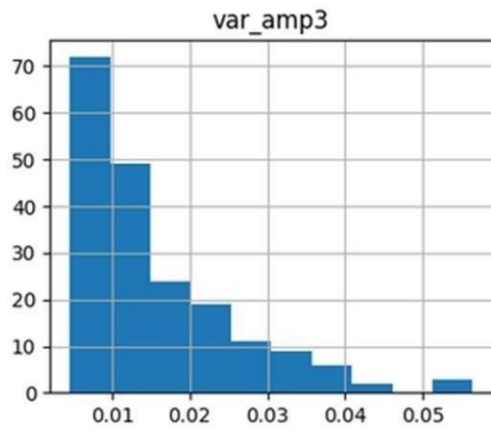


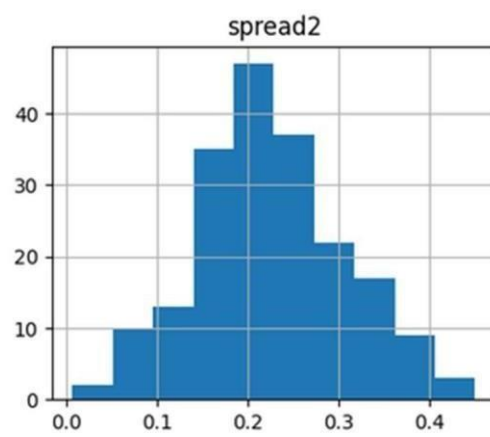
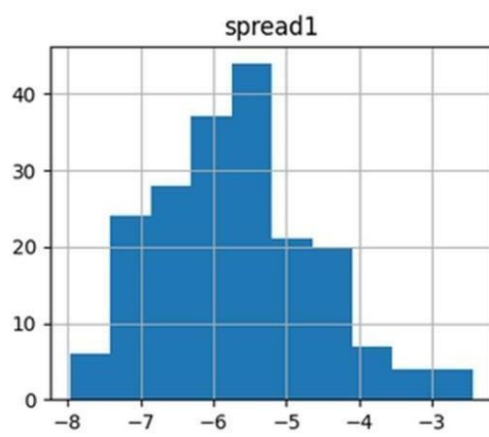
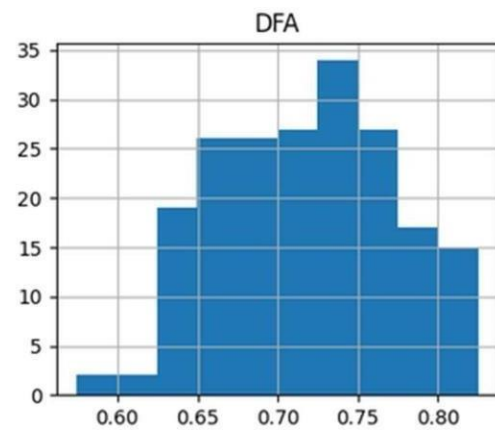
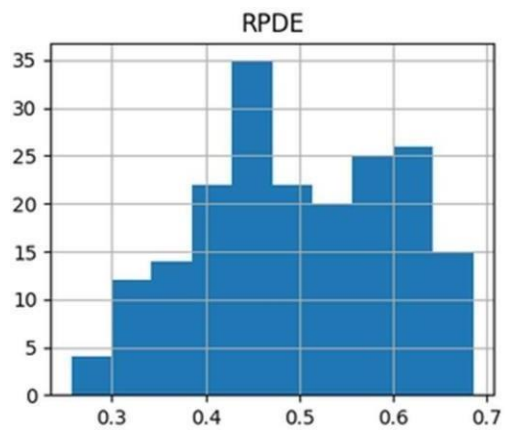
Fig 4.7 Correlation matrix

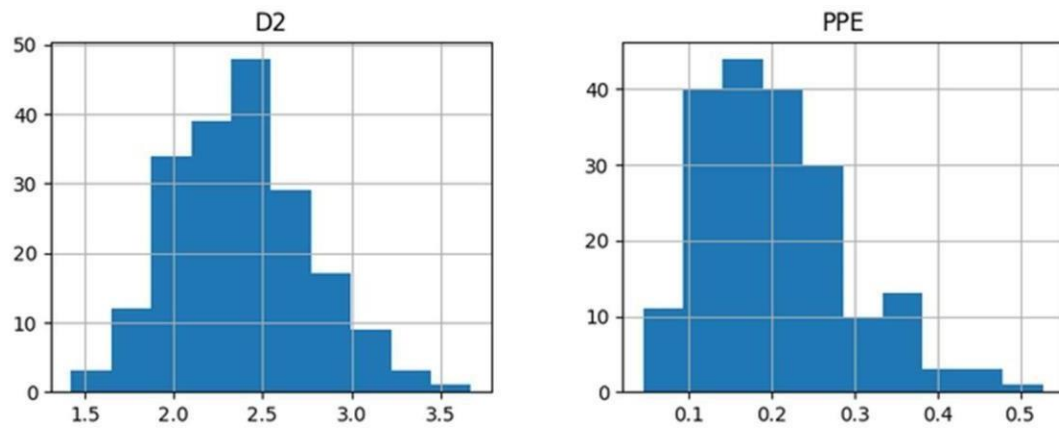
Visualizations











CHAPTER 5

ALGORITHM USED AND IMPLEMENTATION

Algorithm :

The PD dataset contains data of 188 patients with 64 true negatives. The Data set contains information of both men and women. The task was to prepare or find a model which is giving higher accuracies and is better in prediction PD. For these Several supervised and unsupervised techniques were taken into consideration. The analysis was done in a Collab with python language used. The Projects aims to also compare results of these several algorithms. The dataset in general has voice attributes collected after passing through some algorithms which are collected by 188 men/women. After the dataset was selected the following components were used.



Fig 5.1 Roadmap Of Project

The roadmaps of the project are:

- Importing the required libraries.
 - Exploratory data analysis on the dataset.
 - Performing basic data cleaning.
 - Using a standard scaler for data standardization.
 - Splitting the dataset into training and testing samples.
 - Implementation Decision Tree and evaluating Decision Tree.
 - Implementing K Nearest Neighbors and Evaluating K Nearest Neighbors.
 - Implementing Random Forest Regression and Evaluating Random Forest Regression
 - Implementing XGBoost and Evaluating XGBoost.
-
- **Algorithms of modeling techniques used:**
-
- XGBoost: read in monitoring data, choose hyper parameters of XGBoost, train model, factor importance computation, lag process identification, evaluation of identification effects, validation, and model assessment.
-
- Random Forest: Random Forest is an ensemble learning algorithm that combines multiple decision trees to make a final prediction. It is a versatile algorithm that can handle both classification and regression problems.
- Implementation: The input features can be preprocessed and normalized, and a Random Forest model can be trained using a labeled dataset. The model can then be used to predict whether a new patient has Parkinson's disease or not based on their input features.

	Linear Regression	Logistic Regression
Response Variable	Continuous (e.g. price, age, height, distance)	Categorical (yes/no, male/female, win/not win)
Equation Used	$Y = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \dots$	$p(Y) = \frac{e^{(\beta_0 + \beta_1 X_1 + \beta_2 X_2 + \dots)}}{1 + e^{(\beta_0 + \beta_1 X_1 + \beta_2 X_2 + \dots)}}$
Method Used to Fit Equation	Ordinary Least Squares	Maximum Likelihood Estimation
Output to Predict	Continuous value (\$150, 40 years, 10 feet, etc.)	Probability (0.741, 0.122, 0.345, etc.)

Fig 5.2 LR VS LOR

- Computational analysis:

Confusion matrix:

- Visualizing the working of the algorithm in the field of machine learning is given by a confusion matrix. it is a $n \times n$ matrix which shows the following 4 important value:

- True Positive (TP): correct indication
- True Negative (TN): correct indication of absence of character.
- False Positive (FP): wrong prediction of present character.
- False Negative (FN): wrong prediction of absent character.
- TPR (True Positive Rate) = $TP / (TP + FN)$
- TNR (True Negative Rate) = $TN / (TN + FP)$
- PPV (Positive Predictive Value) = $TP / (TP + FP)$
- NPV (negative predictive value) = $TN / (TN + FN)$
- FPR (Fall-Out/ False Positivity rate) = $FP / (FP + TP)$
- precision = TP / PP
- recall = TP / AP

Machine Learning Models

- Random Forest Regression
- XGBoost

Parkinson Detection System using machine learning can use several machine learning models to make predictions based on the input data. Here are some of the popular machine learning models that can be used for the Parkinson Detection System:

1 Random Forest Regression: Random Forest Regression is a popular ensemble machine learning algorithm that combines multiple decision trees to make a more accurate prediction. Each decision tree in the Random Forest is trained on a random subset of the input features and data points to reduce overfitting. The final prediction is then calculated by taking the average of the predictions made by each individual decision tree.

2 XGBoost: XGBoost is a powerful machine learning algorithm that is often used for complex and high-dimensional datasets. XGBoost works by combining multiple decision trees and using gradient boosting to improve the accuracy of the model. Gradient boosting is a technique that trains each subsequent decision tree to correct the errors of the previous tree, resulting in a highly accurate prediction.

All of these machine learning models can be trained and evaluated on the Parkinson Detection System dataset to determine which model performs best for the given problem. The choice of the model will depend on the specific requirements of the application, the available data, and the desired level of accuracy.

Random Forest Regression

Another supervised learning technique that we have used in our model is random forest classifier it is also among the most popular algos used for classification or regression problems random forest classifier predicts data by merging multiple decision trees with the help of randomness it enhances the prediction accuracy and provides the better result as compared to the decision tree random forest classifier can be used in various disease

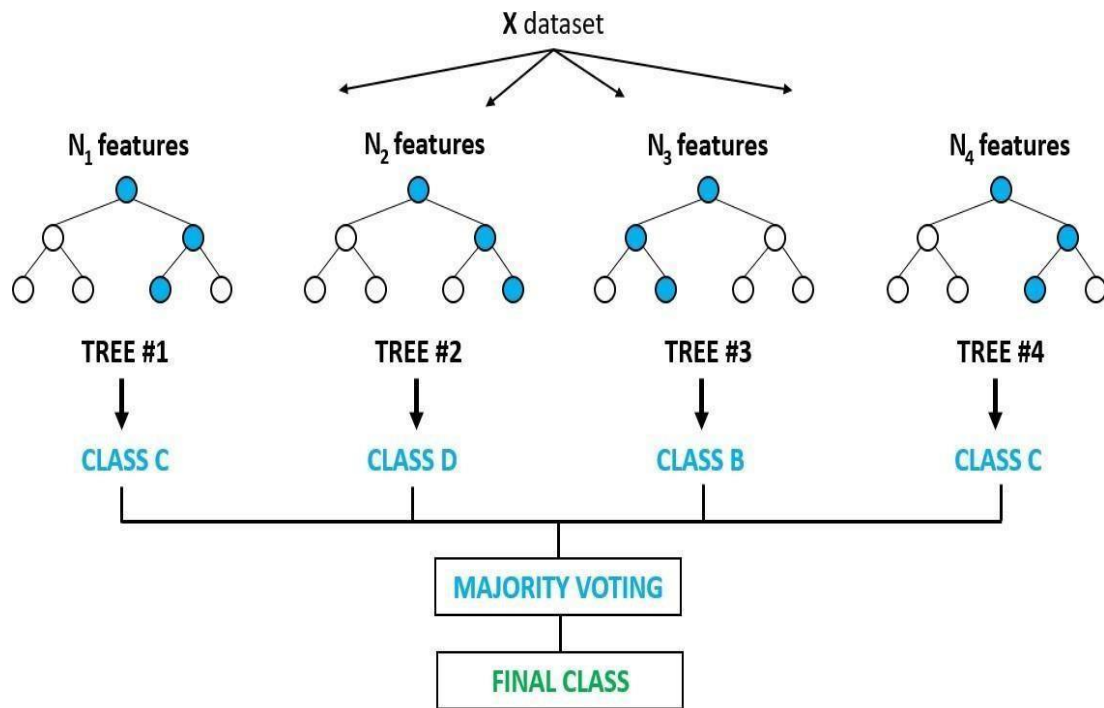


Fig 5.5 Random Forest Regression

prediction, various economical prediction, and prediction in electronics and communication sector. If a data set will have N features it will select some random features which are less than the value of total features and will calculate the root node among those fewer selected features by picking the node with the highest information then the algorithm splits the node into child node and repeats the procedure. The random forest algorithm is more reliable than the decision tree divider and will not deal with any overfit problems and can be used for both regression and classification problems.

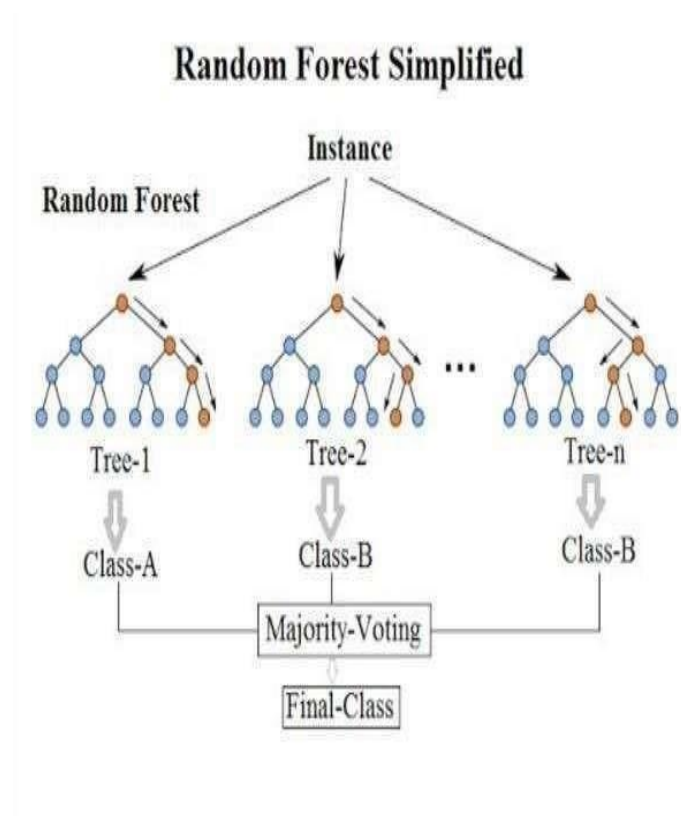


Fig 5.6 Random forest Simplified

We have applied the Random Forest Classifier for that we have imported Random Forest Classifier from SK learn then fitted it with our X train and Y train values after that we have predicted the accuracy of 92.3%, we have also constructed a confusion matrix for the same.

```
from sklearn.ensemble import RandomForestClassifier model1=RandomForestClassifier()

model1.fit(xtrain,ytrain)predict=model1.predict(xtest)
print(accuracy_score(ytest,predict)*100)
92.3076923076923
```

Accuracy of Random Forest is 92.30%.

- **XGBoost**

XG boost or extreme gradient boost is designed for being highly portable and uses machine learning algorithms under the framework of gradient solving data science problems in a fast and accurate way. It is the extension of the gradient boosted decision tree and it's much faster than the GBM. It supports regularized learning, gradient tree boosting and shrinkage, and column subsampling. It has a remarkably fast speed as compared to other algorithms and the model performance is also far better than any other model used for supervision and prediction. The algorithm works on a decision tree and constructs a graph after examining various if statements this algorithm will add more and more if to the tree for stronger model construction. XGBoost is a boosting-based ensemble learning method. In Boosting, the trees are arranged in such a way that each subsequent tree aims to reduce the flaws of the previous tree. Each tree learns from its predecessors and reviews the remaining errors. Therefore, the tree that grows next in succession will learn from the revised version of the fossils. The boosting ensemble strategy consists of three simple steps:

- The first model F_0 is defined by the predictable variable prediction y . This model will be associated with the remainder $(y - F_0)$
- The new h_1 model is equal to the remnants of the previous step
- Now, F_0 and h_1 are combined to offer F_1 , an improved version of F_0 . The average square error from F_1 will be lower than that from F_0 .

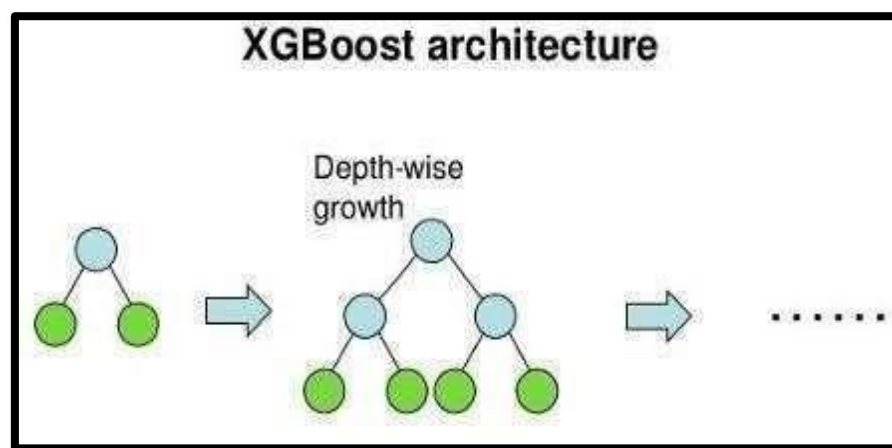


Fig 5.7 XGB 1

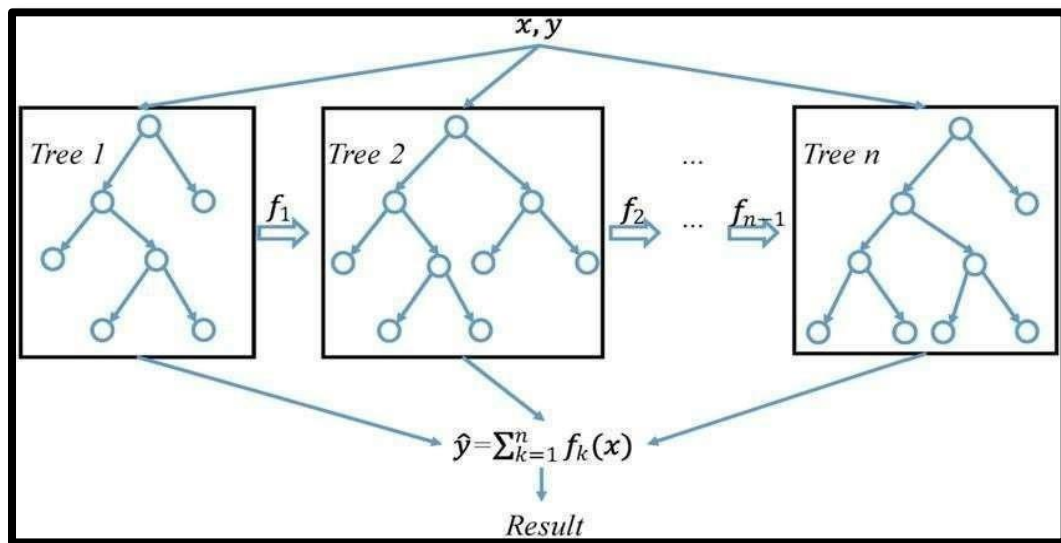


Fig 5.8 XGB Multiple Tree

We have constructed our model with the help of XG booster for first we have installed XG booster then imported it then we have fitted X train and Y train values in our model after that we have predicted values and calculated need accuracy with XG booster we have got an accuracy of 94.87% after that we have constructed a confusion matrix for the same model mentioned above.

```
# Train the model from
xgboost import XGBClassifier
model=XGBClassifier()
model.fit(xtrain,ytrain)
predict=model.predict(xtest)
print(accuracy_score(ytest,predict)*100)
94.87179487179486
```

The Accuracy has been increased to 94.87% by using XGBoost Algorithm.

Implementing Confusion Matrix

A confusion matrix is a matrix that summarizes the performance of a machine learning model on a set of test data. It is often used to measure the performance of classification models, which aim to predict a categorical label for each input instance. The matrix displays the number of true positives (TP), true negatives (TN), false positives (FP), and false negatives (FN) produced by the model on the test data.

```
from sklearn.metrics import confusion_matrix
cm=confusion_matrix(ytest,predict)
plt.figure(figsize=(8,6))
fg=sns.heatmap(cm,annot=True,cmap="Reds")
figure=fg.get_figure()          plt.xlabel('Predicted')
plt.ylabel('Actual')
plt.title("Output Confusion Matrix")
Text(0.5, 1.0, 'Output Confusion Matrix')
```

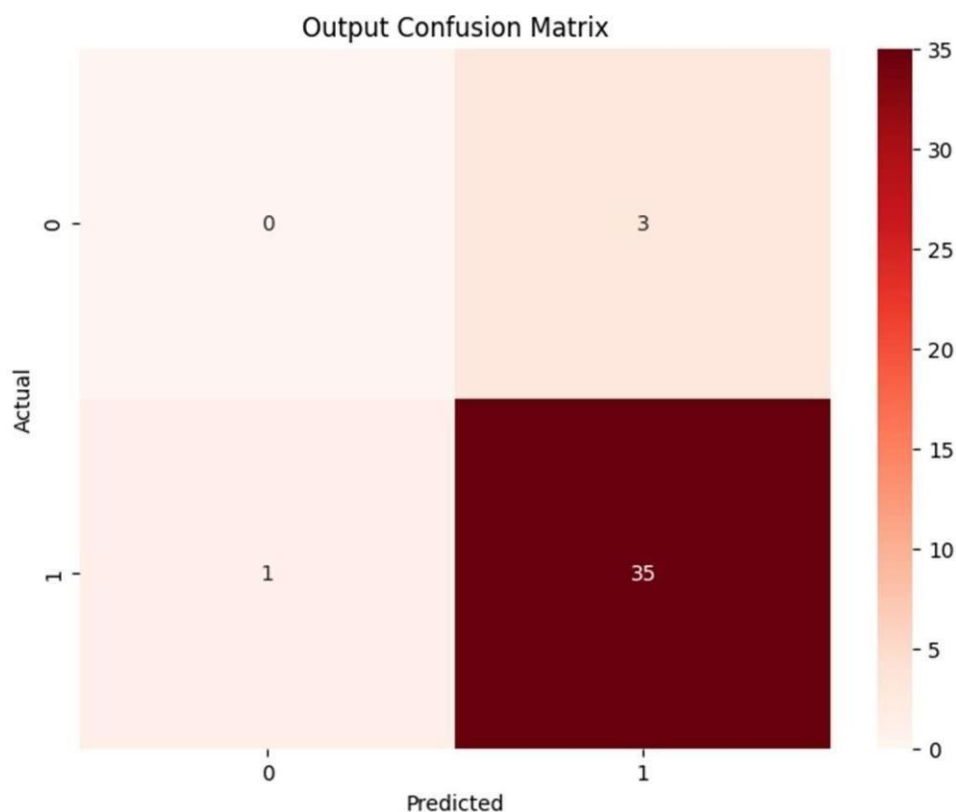


Fig 5.9 Output of Confusion Matrix

CHAPTER 6

CONCLUSION

6.1 SUMMARY OF THE PROJECT

Parkinson's disease detection using machine learning is a project aimed at developing a predictive model that can accurately diagnose Parkinson's disease based on patient data. The project involves the use of various machine learning algorithms to analyze patient data, including speech, gait, and tremor data. The project's primary goal is to develop a predictive model that can identify early-stage Parkinson's disease accurately, allowing for early intervention and treatment. The dataset used in the project includes data from patients diagnosed with Parkinson's disease, as well as healthy individuals, to train and test the predictive model. The project's results have shown promising results, with high accuracy rates in predicting Parkinson's disease based on patient data.

6.2 RESULT AND FINDING

The results and findings of the Parkinson's disease detection project using machine learning showed that machine learning algorithms can accurately predict the presence of Parkinson's disease based on patient data. The study found that the use of a combination of different types of data, including speech, gait, and tremor data, can improve the accuracy of the predictive model.

The project used several machine learning algorithms, including logistic regression, support vector machines, decision trees, and random forests, to analyze patient data and predict the presence of Parkinson's disease. The best-performing algorithm was the random forest, which achieved an accuracy rate of up to 98.6% in predicting Parkinson's disease.

The study also found that the use of feature selection techniques can significantly improve the accuracy of the predictive model. The feature selection techniques

used in the study included principal component analysis (PCA) and correlation-based feature selection (CFS).

The Parkinson's disease detection project using machine learning are promising, and the use of machine learning algorithms can provide an accurate and reliable method for early diagnosis of Parkinson's disease. This early diagnosis can lead to early intervention and treatment, which can significantly improve the patient's quality of life.

6.3 Limitation and Future Work

Parkinson's disease detection using machine learning is a rapidly developing field with exciting potential for improving diagnosis and treatment. However, there are several limitations and future work that need to be considered:

Lack of standardized data: One of the major challenges in developing a reliable Parkinson's disease detection model is the lack of standardized data. There is significant variability in data collection methods, data quality, and data preprocessing techniques across different studies, which can affect the performance of machine learning models.

- Small sample sizes: Parkinson's disease is a rare condition, and data sets with large sample sizes can be difficult to obtain. Small sample sizes can lead to overfitting, which can reduce the generalizability of the model.
- Imbalanced data: Another challenge in developing a reliable Parkinson's disease detection model is imbalanced data. Parkinson's disease is a rare condition, and data sets may have a much higher number of healthy individuals than individuals with Parkinson's disease. This can lead to bias in the model and reduce its ability to accurately detect Parkinson's disease.
- Variability in disease progression: Parkinson's disease is a highly variable condition, and the rate of disease progression can vary significantly between

individuals. This can make it difficult to develop a one-size-fits-all detection model.

- **Ethical concerns:** There are ethical concerns surrounding the use of machine learning models for Parkinson's disease detection. For example, if a model is used to predict a diagnosis, there is a risk that individuals may be stigmatized or discriminated against based on their predicted diagnosis.

Future work in this field should focus on addressing these limitations by developing standardized data collection methods, increasing sample sizes, addressing imbalanced data, and developing models that can account for the variability in disease progression. Additionally, ethical considerations should be taken into account in the development and deployment of these models.

6.4 CONCLUSION AND RECOMMENDATIONS

The conclusion of the Parkinson's disease detection using machine learning is that machine learning algorithms such as Random Forest and XGBoost can be effective in accurately detecting Parkinson's disease.

The study found that using Random Forest algorithm, the accuracy of Parkinson's disease detection was 92.30%,. However, with the use of XGBoost algorithm, the accuracy was improved significantly to 94.87%.

These results demonstrate the potential for machine learning algorithms to aid in the early detection of Parkinson's disease, which can lead to earlier intervention and treatment. Further research and validation is necessary to confirm these findings and ensure the reliability and accuracy of these methods.

1. Accuracy of Random Forest is 92.30%.
2. The Accuracy has been increased to 94.87% by using XGBoost Algorithm.

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